복사전달 특강

Special Topics in Radiative Transfer

Lecture 3

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[Electron-Ion Collisional Processes]

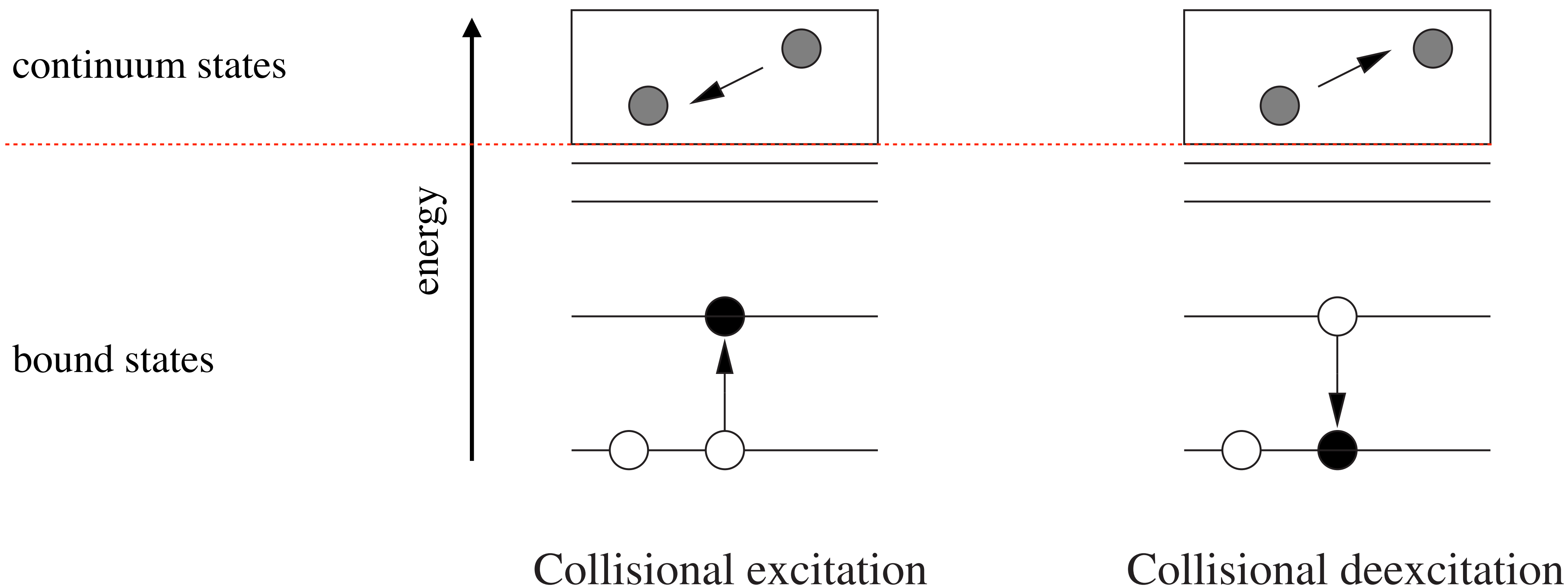
- To predict the emergent spectra of astrophysical plasmas, we need to understand the details of how excited atomic levels are populated. For the most part, it involves the study of electron-ion collisional processes in gas.
- Each electron-ion collisional process is accompanied by a quantum mechanical inverse, which can be viewed as the same process time-reversed.

- There are essentially four (+1) key electron-ion collisional processes.
  - Collisional Excitation / Deexcitation
  - Collisional Ionization / Three-Body Recombination
  - Radiative Recombination / Photoionization
  - Dielectronic Recombination (Capture) / Autoionization
  - Charge Exchange

- Softwares & Data
  AUTOSTRUCTURE (https://amdpp.phys.strath.ac.uk/autos/)
  NIST (https://www.nist.gov/pml/atomic-spectroscopy-databases)

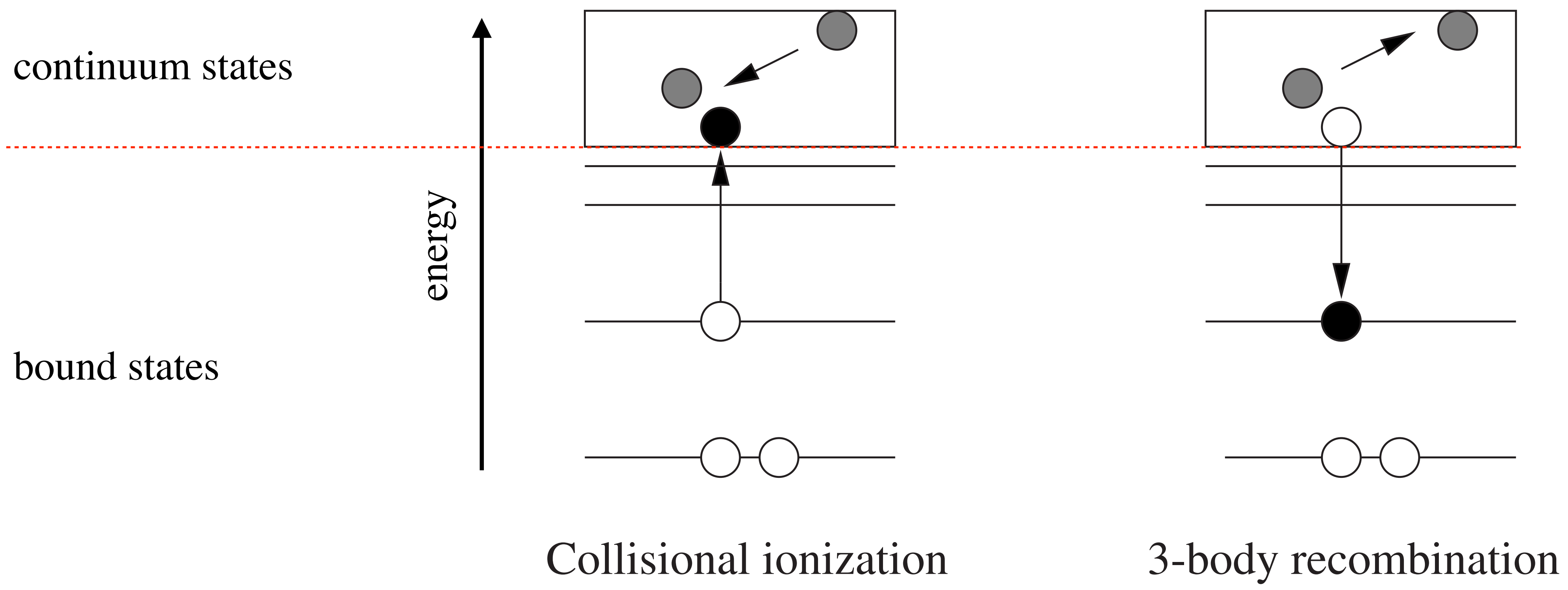
(1) Collisional Excitation / Deexcitation

- In collisional excitation, the interaction between a passing electron in a continuum state and a bound electron in a discrete state result in the excitation of the bound electron to a higher energy discrete level.
- To conserve energy, the colliding electron gives up a fraction of its energy and thus "falls" into a lower continuum state.
- The inverse process is collisional deexcitaiton, where a passing electron interacting with an excited atom actually gains energy as a result of the collisions. (No photon emission occurs.)



(2) Collisional Ionization / Three-Body Recombination

- Collisional ionization is similar to collisional excitation, except that in this case, the final state of the initially bound electron is also a continuum state.
- The inverse process is 3-body recombination. Here, two initially free electrons interact with the ion in the same collision. One of the two gets captured into a bound discrete level, while the other carries off the excess energy in a higher continuum state.

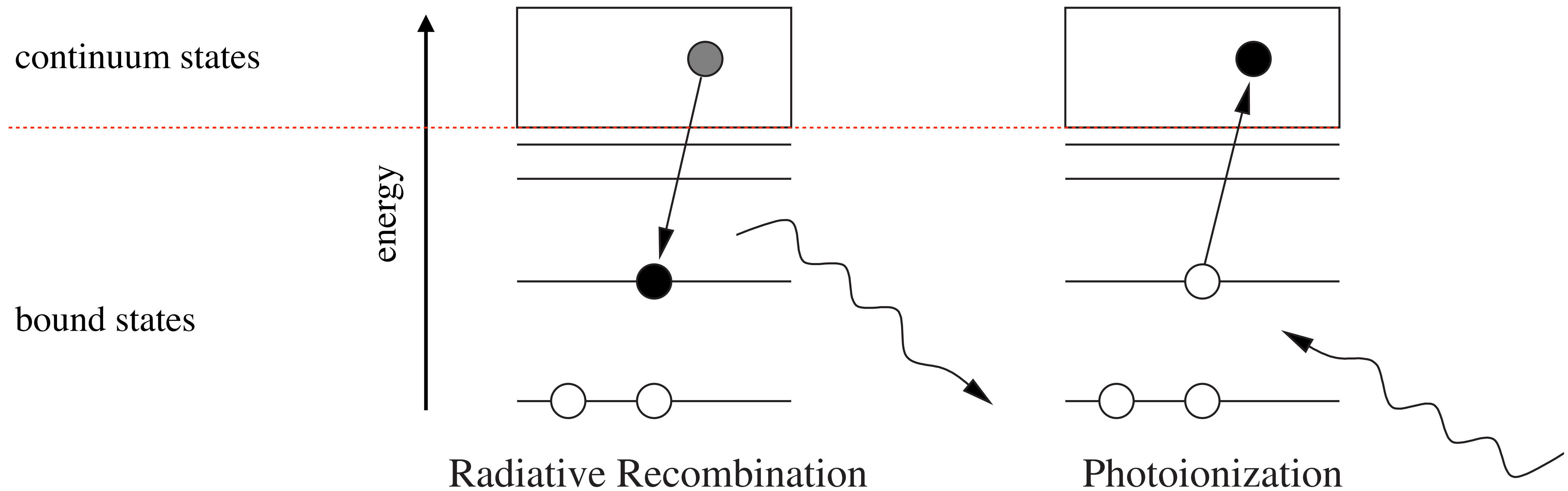


(3) Radiative Recombination / Photoionization

- In radiative recombination a free electron in a continuum state decays into a bound discrete state through the emission of a photon. This is actually a form of a spontaneous emission, similar to the radiative decay between two bound levels.

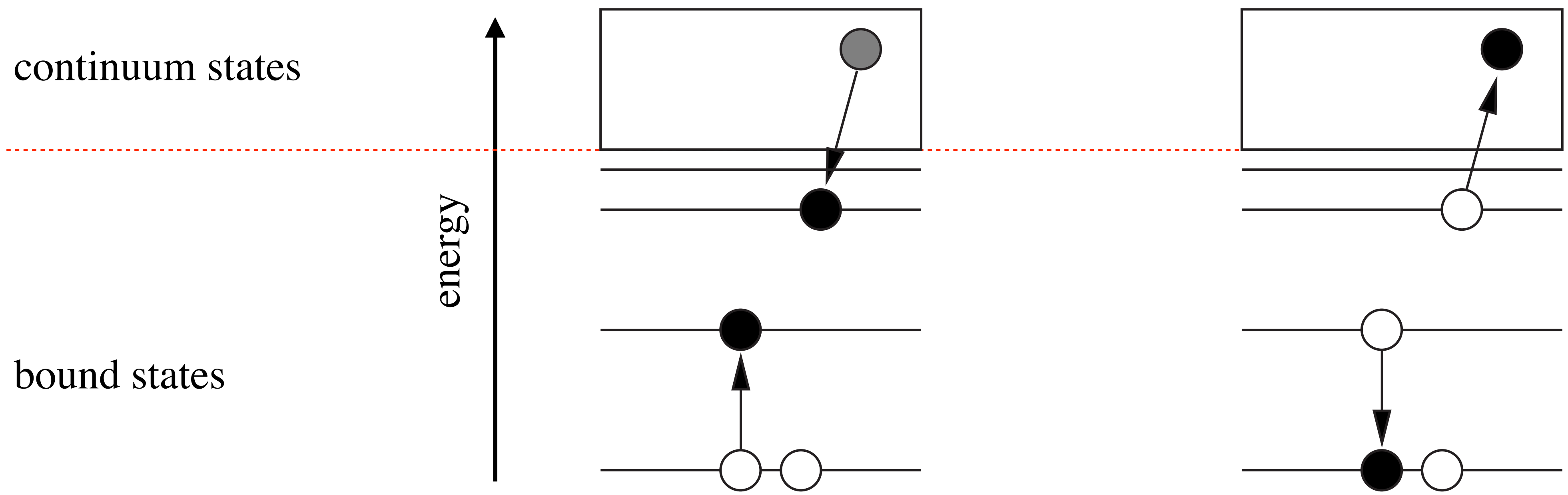
$$\boxed{\begin{array}{rcll} A^{i+} + e^- & \rightarrow & A_*^{(i-1)+} + h\nu & \text{(recombination continuum)} \\ A_*^{(i-1)+} & \rightarrow & A^{(i-1)+} + h\nu_1 + h\nu_2 + h\nu_3 + \cdots & \text{(recombination lines)} \end{array}}$$

- The inverse process is photoionization or bound-free absorption.
  - Interstellar medium (ISM) is transparent to $h\nu$ < 13.6 eV photons, but is very opaque to ionizing photons ($h\nu$ > 13.6 eV). In fact, the ISM does not become transparent until $h\nu$ ~ 1 keV. Sources of ionizing photons include massive, hot young stars, hot white dwarfs, and supernova remnant shocks.



(4) Dielectronic Recombination (Capture) / Autoionization

- Dielectronic recombination (capture) is a resonant radiationless process in which the decay of an electron from a continuum state to a bound state is accompanied by the elevation of a core electron into an excited state. The resulting atom is doubly excited, and it has a total energy above the ionization potential of the initial ion.
- The inverse process is autoionization, where a doubly excited atom decays via the emission of a weakly bound outer electron. If the core excitation is associated with a "hole", in one of the orbitals of an inner shell, this process is usually called Auger decay.



Dielectronic Capture

Autoionization

(5) Charge Exchange

- During the collision of two ionic species, the charge clouds surrounding each interact, and it is possible that an electron is exchanged between them.

- Since, in virtually all diffuse astrophysical plasmas, hydrogen and helium are overwhelmingly the most abundant species, the charge-exchange reactions which are significant to the ionization balance of the plasma are

$$\begin{aligned} A^{(i+1)+} + \mathrm{H}^0 &\rightleftharpoons A^{i+} + \mathrm{H}^+ + \Delta E \\ A^{(i+1)+} + \mathrm{He}^0 &\rightleftharpoons A^{i+} + \mathrm{He}^+ + \Delta E \end{aligned}$$

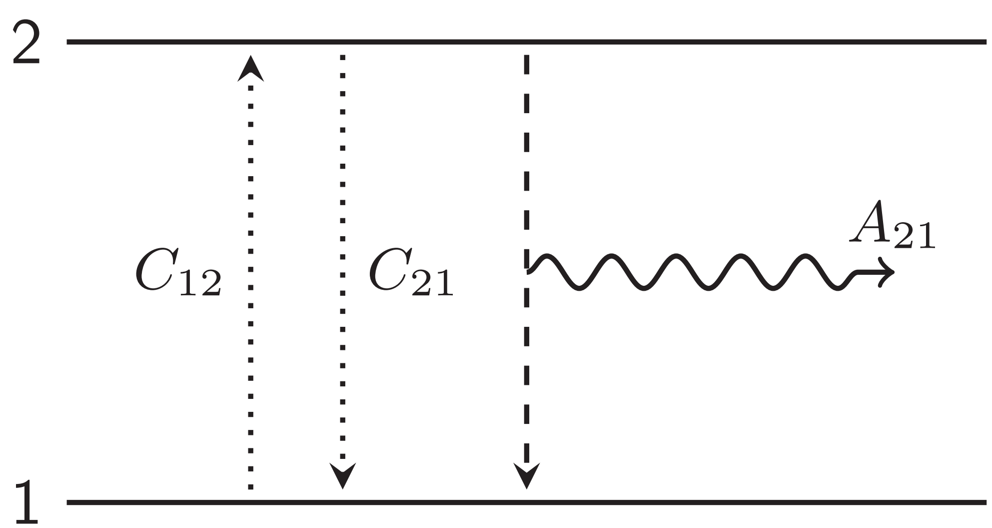
[Ionization Equilibrium]

- **Collisional Ionization Equilibrium (CIE) or coronal equilibrium**
    - dynamic balance at a given temperature between collisional ionization from the ground states of the various atoms and ions, and the process of recombination from the higher ionization stages.
    - In this equilibrium, effectively, all ions are in their ground state.
    - Software: **Chianti** (https://www.chiantidatabase.org/)

- **Photoionization Equilibrium**
    - dynamic balance between photo-ionization and the process of recombination.
    - Softwares
      Cloudy (https://trac.nublado.org/)
      MAPPINGS (https://mappings.anu.edu.au/)
      MOCASSIN (https://github.com/mocassin, https://mocassin.nebulousresearch.org/)
      CMacIonize (https://bwvdnbro.github.io/CMacIonize/)
      XSTAR (https://heasarc.gsfc.nasa.gov/docs/software/xstar/xstar.html)

[Atomic Emission Line Mechanisms]

- **Collisional Line**
  (collisional excitation + spontaneous emission)



Note that collisional de-excitation yields no photons

Collisions with electrons can excite ions (or atoms), moving it from level 1 to 2, or de-excite it, moving it from level 2 to 1, with rate C_{21} and C_{12}, respectively.

An excited ion can also spontaneously emit radiation at the rate of the Einstein A_{21} coefficient.

In low density ISM, collisions are rare, so when the occasional collisional excitation happens, the ion(atom) is much more likely to return to level 1 through the emission of a photon than through collisional de-excitation.

- **Recombination Line (photoionization + recombination / collisional ionization + recombination)**

  Hot stars produce ultraviolet radiation that can ionize hydrogen and other atoms. Once ionized, recombination back to the neutral state produces recombination continuum and lines. Of course, the recombination requires that the ion encounter an electron, which is a slow process in diffuse gas.

  The recombination of a free electron with a proton can occur to any of the energy levels n. For instant, if $n > 1$, the excited hydrogen atom will then decay to lower levels until it reaches the ground state, $n = 1$. This cascade produces a set of protons, the first with an energy corresponding to the potential of the previously unbound electron-proton pair and then the others at fixed values corresponding to the discrete energy jumps between different n.

  In most cases, the Lyman alpha and Balmer lines are recombination lines.

[Collisional Excitation]

- Under the conditions of **very low density** and weak radiation fields,

  - **The vast majority of the atoms reside in the ground state.**

    Collisional excitation timescale $\gg$ *Radiative decay time scale*

    This condition will remain true even if the excited state has a radiative lifetime of several second, which is frequently the case for the forbidden transitions observed in ionized astrophysical plasmas.

  - ***Flux of an emission line*** $\propto$ ***number of collisions*** $\propto$ ***product of the number densities of the two colliding species by the probability that a collision will produce a collisional excitation.***

  - If the energy gap between the ground state and the excited state E_{12} is much larger than the mean energy of the colliding species ($\sim T$), then, because there are few very energetic collisions, relatively few collisional excitations can occur. Therefore, the resulting emission line will be very much weaker than when $E_{12} < kT$.

    This gives us the possibility of measuring temperature from the relative strengths of lines coming from excited levels at different energies above the ground state.

- At high enough densities,
  - The collisional timescales are short.
  - The population in any upper level is set by the balance between collisional excitation, and the collisional de-excitation out of these levels, and are governed by the Boltzmann equilibrium.

Boltzmann equilibrium: $$\frac{n_2}{n_1} = \frac{g_2}{g_1} \exp\left(-\frac{E_{12}}{k_{\rm B}T}\right)$$

statistical weight
$g_i = 2J_i + 1$

- At intermediate densities,
  - The collisional rates and the radiative decay rates are compatible.
  - The intensity of an emission line is determined by both the temperature and the density.
  - If the temperature is known, the density can be determined from the intensity ratio of two such lines.

Collisional Excitation and Deexcitation

- Collisional Rates
  - C_{lu} = number of collisional excitations from state l to state u per sec per particle in state l
  - C_{ul} = number of collisional deexcitations from state l to state u per sec per particle in state l

- Collisional Rate Coefficients for collisions with electrons
  - $k_{lu} = C_{lu}/n_e$
  - $k_{ul} = C_{ul}/n_e$

- Electron collisions (usually the most important ones) causing transitions from state i to state j have transition rates:

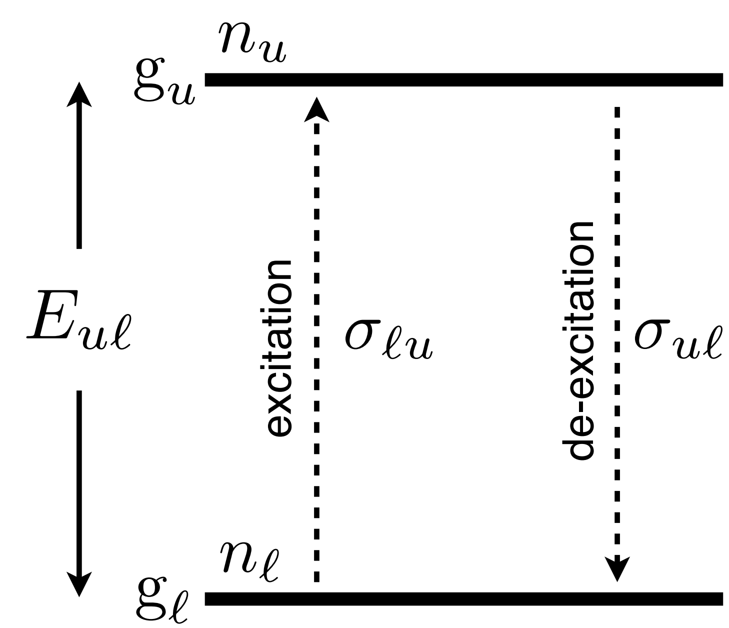
$$\begin{aligned} n_i C_{ij} &= n_i n_e \int_{v_0}^{\infty} \sigma_{ij}(v) v f(v) dv \\ &= n_i n_e k_{ij} \end{aligned}$$

  - Here, n_e is the electron density, $\sigma_{ij}(v)$ the electron collision cross-section, $f(v)$ the normalized velocity distribution (usually Maxwellian), and v_0 the threshold velocity with $(1/2)mv_0^2 = h\nu_0$.

Collisional Rate Coefficients

- **Collisional Rate (Two Level Atom)**
  - The cross section $\sigma_{\ell u}$ for collisional excitation from a lower level ℓ to an upper level u is, in general, inversely proportional to the impact energy (or v^2) above the energy threshold $E_{u\ell}$ and is zero below.
  - The collisional cross section can be expressed in the following form using a dimensionless quantity called the ***collision strength*** $\Omega_{\ell u}$:

$$\boxed{\begin{aligned}\sigma_{\ell u}(v) &= (\pi a_0^2)\left(\frac{hR_{\mathrm{H}}}{\frac{1}{2}m_e v^2}\right)\frac{\Omega_{\ell u}}{\mathrm{g}_\ell} \quad \mathrm{cm}^2 \quad \mathrm{for} \quad \frac{1}{2}m_e v^2 > E_{u\ell} \\ &= \frac{h^2}{4\pi m_e^2 v^2}\frac{\Omega_{\ell u}}{\mathrm{g}_\ell}\end{aligned}}$$



$$\mathrm{or} \quad \sigma_{\ell u}(E) = \frac{h^2}{8\pi m_e E}\frac{\Omega_{\ell u}}{\mathrm{g}_\ell} \qquad \left(E = \frac{1}{2}m_e v^2\right)$$

$$\mathrm{where,} \quad a_0 = \frac{\hbar^2}{m_e e^2} = 5.12 \times 10^{13} \quad \mathrm{cm, \ Bohr\ radius}$$

$$R_{\mathrm{H}} = \frac{m_e e^4}{4\pi\hbar^3} = 109,737 \quad \mathrm{cm}^{-1}, \quad \mathrm{Rydberg\ constant} \quad \left(\hbar = \frac{h}{2\pi}\right)$$

  - The collision strength $\Omega_{\ell u}$ is a function of electron velocity (or energy) but is often approximately constant near the threshold. Here, g_ℓ and g_u are the statistical weights of the lower and upper levels, respectively.

- Advantage of using the collision strength is that (1) it removes the primary energy dependence for most atomic transitions and (2) they have the symmetry between the upper and the lower states.

The principle of detailed balance states that ***in thermodynamic equilibrium each microscopic process is balanced by its inverse.***

$$n_e n_\ell v_\ell \sigma_{\ell u}(v_\ell) f(v_\ell) dv_\ell = n_e n_u v_u \sigma_{u\ell}(v_u) f(v_u) dv_u$$

Here, v_ℓ and v_u are related by $\frac{1}{2}m_e v_\ell^2 = \frac{1}{2}m_e v_u^2 + E_{u\ell}$, and $f(v)$ is a Maxwell velocity distribution of electrons. Using the Boltzmann equation of thermodynamic equilibrium,

$$\frac{n_u}{n_\ell} = \frac{g_u}{g_\ell} \exp\left(-\frac{E_{u\ell}}{kT}\right)$$

we derive the following relation between the cross-sections for excitation and de-excitation are

$$g_\ell v_\ell^2 \sigma_{\ell u}(v_\ell) = g_u v_u^2 \sigma_{u\ell}(v_u) \quad \text{Here, } \frac{1}{2}m_e v_\ell^2 = \frac{1}{2}m_e v_u^2 + E_{u\ell} \longrightarrow g_\ell \cdot (E + E_{u\ell}) \cdot \sigma_{\ell u}(E + E_{u\ell}) = g_u \cdot E \cdot \sigma_{u\ell}(E)$$

where $E = \frac{1}{2}m_e v_u^2$

and the symmetry of the collision strength between levels.

$$\Omega_{\ell u} = \Omega_{u\ell} \qquad \text{more precisely} \quad \Omega_{\ell u}(E + E_{u\ell}) = \Omega_{u\ell}(E)$$

These two relations were derived in the TE condition. However, ***the cross-sections are independent on the assumption, and thus the above relations should be always satisfied.***

- Collisional excitation and de-excitation rates

The ***collisional de-excitation rate per unit volume per unit time, which is thermally averaged,*** is

$$\left(\frac{dn_\ell}{dt}\right)_{u\to\ell} = n_e n_u \int_0^\infty v\sigma_{u\ell}(v) f(v)dv = n_e n_u k_{u\ell} \quad [\mathrm{cm}^{-3}\ \mathrm{s}^{-1}]$$

$$\boxed{k_{u\ell} \equiv \langle \sigma v \rangle_{u\to\ell}}$$

$$\begin{aligned} k_{u\ell} &= \int_0^\infty v\sigma_{u\ell}(v) f(v)dv \\ &= \left(\frac{2\pi\hbar^4}{k_{\rm B} m_e^3}\right)^{1/2} T^{-1/2} \frac{\langle \Omega_{u\ell}\rangle}{g_u} \\ &= \frac{8.62942\times 10^{-6}}{T^{1/2}} \frac{\langle \Omega_{u\ell}\rangle}{g_u} \quad [\mathrm{cm}^3\ \mathrm{s}^{-1}], \end{aligned}$$

effective collision strength:

$$\langle \Omega_{u\ell}\rangle \equiv \int_0^\infty \Omega_{u\ell}(E) e^{-E/k_{\rm B}T} d(E/k_{\rm B}T)$$

and the ***collisional excitation rate per unit volume per unit time*** is

$$\left(\frac{dn_u}{dt}\right)_{\ell\to u} = n_e n_\ell k_{\ell u}$$

$$\boxed{k_{\ell u} \equiv \langle \sigma v \rangle_{\ell\to u}}$$

$$k_{\ell u} = \int_{v_{\rm min}}^\infty v\sigma_{\ell u}(v) f(v)dv \quad \text{Here, } \frac{1}{2} m_e v_{\rm min}^2 = E_{u\ell}$$

$$= \left(\frac{2\pi\hbar^4}{k_{\rm B} m_e^3}\right)^{1/2} T^{-1/2} \frac{\langle \Omega_{u\ell}\rangle}{g_\ell} \exp\left(-\frac{E_{u\ell}}{k_{\rm B}T}\right)$$

Here, $k_{\ell u}$ and $k_{u\ell}$ are the **collisional rate coefficient for excitation and de-excitation** coefficients in units of cm^3 s^{-1}, respectively. We also note that ***the rate coefficients for collisional excitation and de-excitation are related by***

$$\boxed{\frac{k_{\ell u}}{k_{u\ell}} = \frac{g_u}{g_\ell} \exp\left(-\frac{E_{u\ell}}{k_{\rm B}T}\right)} \qquad k_{\ell u} = \frac{g_u}{g_\ell} k_{u\ell} \exp\left(-\frac{E_{u\ell}}{k_{\rm B}T}\right) \qquad \boxed{\langle \sigma v \rangle_{\ell\to u} = \frac{g_u}{g_\ell} \langle \sigma v \rangle_{u\to\ell} \exp\left(-\frac{E_{u\ell}}{k_{\rm B}T}\right)}$$

Sum rule for collision strengths

- Quantum mechanical sum rule for collision strengths for the case where one term consists of a singlet ($S = 0$ or $L = 0$) and the second consists of a multiplet: the collision strength of each fine structure level J is related to the total collision strength of the multiplet by

$$\boxed{\Omega_{(SLJ,\ S'L'J')} = \frac{(2J'+1)}{(2S'+1)(2L'+1)}\Omega_{(SL,\ S'L')}}$$

  Here, $(2J'+1)$ is the statistical weight of an individual level in the multiplet, and $(2S'+1)(2L'+1)$ is the statistical weight of the multiplet term.

  We can regard the collision strength of the term as "shared" amongst these levels in proportion to the statistical weights of the individual levels $(g_J = 2J+1)$.

- ***The flux ratio between the lines in a multiplet is proportional to the ratio of their collision strengths, in a low density medium.*** Then, the flux ratio is determined by the ratio of their statistical weights.
  - C-like ions ($1s^2 2s^2 2p^2 \rightarrow 1s^2 2s^2 2p^2$) forbidden or inter combination transitions.

    ground states (triplet) - $^3P_0 : {}^3P_1 : {}^3P_2 = 1 : 3 : 5$

    excited states (singlets) - $^1D_2$, $^1S_1$
  - Li-like ions ($1s^2 2s^1 \rightarrow 1s^2 2p^1$) resonance transitions

    ground state (singlet) - $^2S_{1/2}$

    excited states (doublet) - $^2P_{3/2} : {}^2P_{1/2} = 2 : 1$

Collisionally-Excited Emission Line

- Emission line flux
  - **In the low density limit**, the collisional rate between atoms and electrons is much slower than the (spontaneous) radiative de-excitation rate of the excited level. Thus, we can balance the collisional feeding into level u by the rate of radiative transition back down to level ℓ. The level population is determined by

$$n_e n_\ell k_{\ell u} = A_{u\ell} n_u \qquad \frac{n_u}{n_\ell} = \frac{n_e k_{\ell u}}{A_{u\ell}} = \frac{n_e}{A_{u\ell}} \beta \frac{\langle \Omega_{u\ell} \rangle}{g_\ell} T^{-1/2} \exp\left(-\frac{E_{u\ell}}{kT}\right)$$

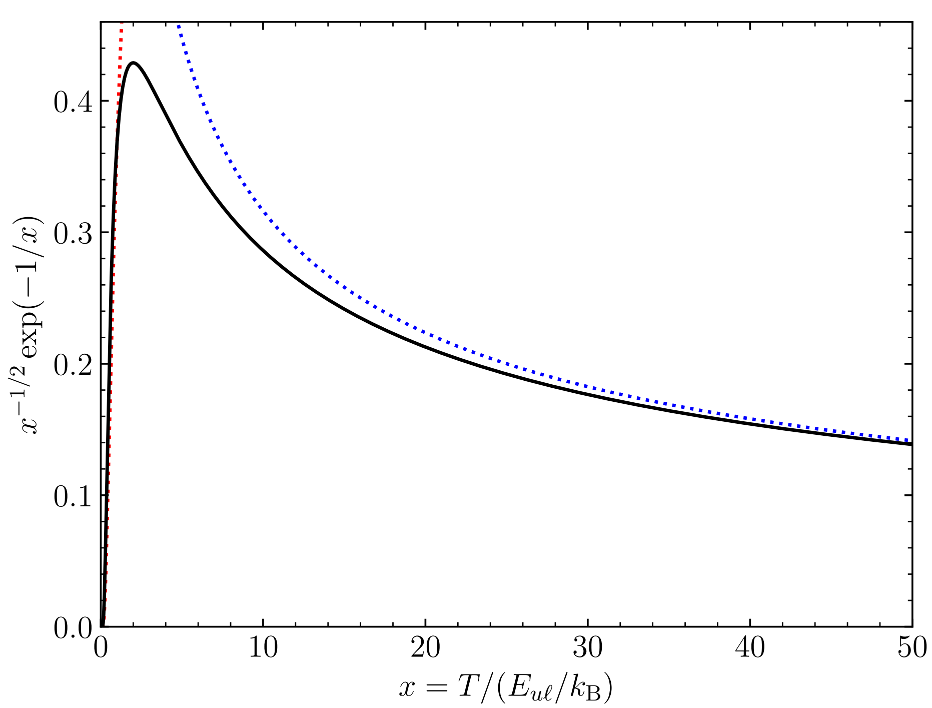
where $A_{u\ell}$ is the Einstein coefficient for spontaneous emission. The line emissivity is given by

$$\begin{aligned} 4\pi j_{u\ell} &= E_{u\ell} A_{u\ell} n_u = E_{u\ell} n_e n_\ell k_{\ell u} \\ &= n_e n_\ell E_{u\ell} \frac{8.62942 \times 10^{-6}}{T^{1/2}} \frac{\langle \Omega_{u\ell} \rangle}{g_\ell} \exp\left(-\frac{E_{u\ell}}{kT}\right) \quad [\mathrm{erg\ cm^{-3}\ s^{-1}}] \\ &\simeq \beta \chi n_e^2 E_{\ell u} T^{-1/2} \frac{\langle \Omega_{u\ell} \rangle}{g_\ell} \exp\left(-\frac{E_{u\ell}}{kT}\right) \end{aligned}$$

Here,

$$\beta = \left(\frac{2\pi\hbar^4}{k m_e^2}\right)^{1/2} = 8.62942 \times 10^{-6}$$

$$\chi = n_\ell / n_e$$



For low temperature, the exponential term dominates because few electrons have energy above the threshold for collisional excitation, so that the line rapidly fades with decreasing temperature.

At high temperature, the $T^{-1/2}$ term controls the cooling rate, so the line fades slowly with increasing temperature.

- **In high-density limit**, the level population are set by the Boltzmann equilibrium, and the line emissivity is

$$\frac{n_u}{n_\ell} = \frac{g_u}{g_\ell} \exp\left(-\frac{E_{u\ell}}{kT}\right)$$

$$\boxed{\begin{aligned} 4\pi j_{u\ell} &= E_{\ell u} A_{u\ell} n_u \\ &= n_\ell E_{\ell u} A_{u\ell} \frac{g_u}{g_\ell} \exp\left(-\frac{E_{\ell u}}{kT}\right) \end{aligned}}$$

$$\simeq \chi n_e E_{\ell u} A_{u\ell} \frac{g_u}{g_\ell} \exp\left(-\frac{E_{\ell u}}{kT}\right)$$

$$\rightarrow \quad n_\ell E_{\ell u} A_{u\ell} \frac{g_u}{g_\ell} \quad \text{as} \quad kT \gg E_{\ell u}$$

Here, **the line flux scales as n_e rather than n_e^2, but the line flux tends to be a constant value at high temperature.**

- **Critical density** is defined as **the density where the radiative depopulation rate matches the collisional de-excitation for the excited state.**

$$A_{u\ell} n_u = n_e n_u k_{u\ell}$$

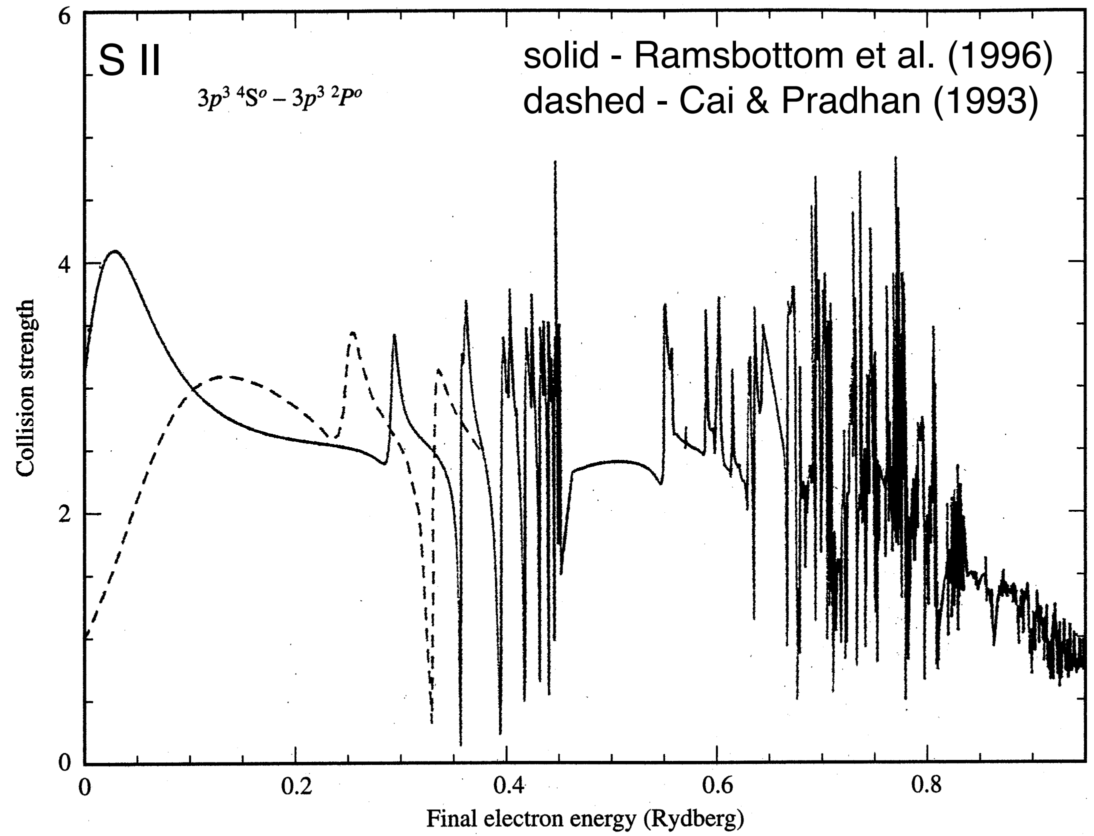
$$n_{\rm crit} = \frac{A_{u\ell}}{k_{u\ell}}$$

$$\rightarrow \quad \begin{aligned} n_{\rm crit} &= A_{u\ell} \frac{g_u}{\beta \langle \Omega_{u\ell} \rangle} T^{1/2} \\ &= 1.2 \times 10^3 \frac{A_{u\ell}}{10^{-4}\,{\rm s}^{-1}} \frac{g_u}{\langle \Omega_{u\ell} \rangle} \left(\frac{T}{10^4\,{\rm K}}\right)^{1/2} \quad [{\rm cm}^{-3}] \end{aligned}$$

- At densities higher than the critical density, collisional de-excitation becomes significant, and the forbidden lines will be weaker as the density increases.

At around the critical density, the "line emissivity vs density" plotted in log-log scale changes slope from +2 to +1.

- Collision Strength
  - Quantum mechanical calculations show that (1) the resonance structure in the collision strengths is important and (2) the collision strength increases with energy for neutral species.



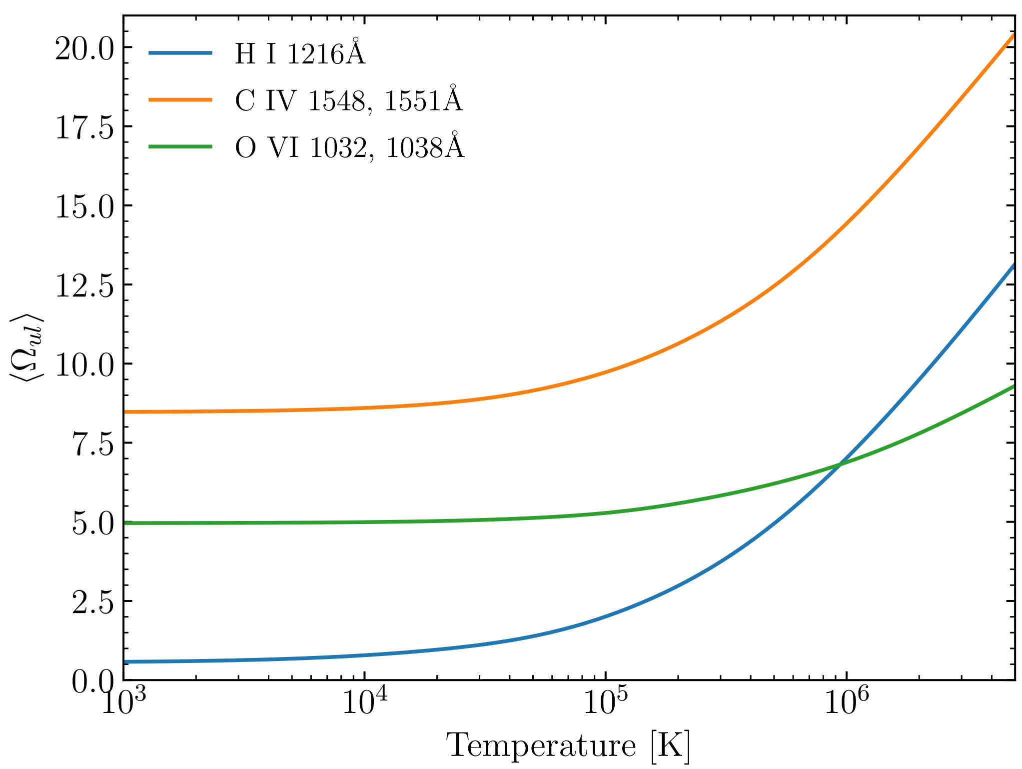
- Effective Collision Strength

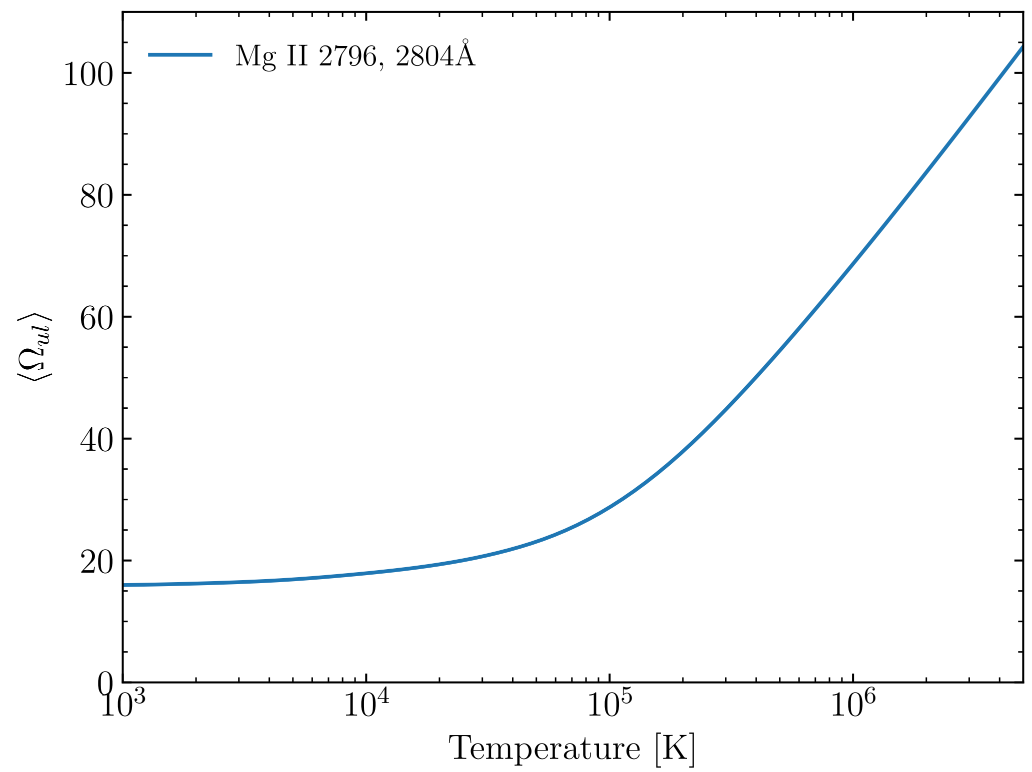
The ***effective collision strength*** has a value in a range of

$$10^{-2} < \langle \Omega_{u\ell} \rangle < 10$$

$$\langle \Omega_{u\ell} \rangle = \int_0^\infty \Omega_{u\ell}(E) e^{-E/k_{\rm B}T} d(E/k_{\rm B}T)$$

Refer to Tables F.1 - F.5 in [Draine]
and the CHIANT atomic database https://www.chiantidatabase.org/





- As can be seen in the Tables and the formula, **collisional de-excitation is negligible** for resonance and most forbidden lines in the ISM.

Ion	ℓ	u	E_ℓ/k (K)	E_u/k (K)	$\lambda_{u\ell}$ (μm)	$n_{\rm H,crit}(u)$ $T=100\,$K (cm^{-3})	$T=5000\,$K (cm^{-3})
C II	^{2}P$^o_{1/2}$	^{2}P$^o_{3/2}$	0	91.21	157.74	2.0×10^3	1.5×10^3
C I	^{3}P$_0$	^{3}P$_1$	0	23.60	609.7	620	160
	^{3}P$_1$	^{3}P$_2$	23.60	62.44	370.37	720	150
O I	^{3}P$_2$	^{3}P$_1$	0	227.71	63.185	2.5×10^5	4.9×10^4
	^{3}P$_1$	^{3}P$_0$	227.71	326.57	145.53	2.3×10^4	8.4×10^3
Si II	^{2}P$^o_{1/2}$	^{2}P$^o_{3/2}$	0	413.28	34.814	1.0×10^5	1.1×10^4
Si I	^{3}P$_0$	^{3}P$_1$	0	110.95	129.68	4.8×10^4	2.7×10^4
	^{3}P$_1$	^{3}P$_2$	110.95	321.07	68.473	9.9×10^4	3.5×10^4

Table 17.1 in [Draine]

- However, it is not true for the 21 cm hyperfine structure line of hydrogen.
  - The critical density for 21cm line is

$$n_{\rm crit} \sim 10^{-3}\,(T/100\,{\rm K})^{-1/2}\ \ [{\rm cm}^{-3}]$$

$$A_{u\ell} = 2.88\times10^{-15}\ [{\rm s}^{-1}]$$

  - The hyperfine levels are thus essentially in collisional equilibrium in the CNM.

The collisional strengths and other atomic data are available in the CHIANTI atomic database (https://www.chiantidatabase.org/).

Collision strengths at T = 10^4 K

Table 4.1 in The Interstellar Medium [Lequeux]

Ion	Transition l–u	λ µm	A_{ul} s^{-1}	Ω_{ul}	n_{crit} cm^{-3}
C I	^{3}P$_0$–^{3}P$_1$	609.1354	7.93×10^{-8}	–	(500)
	^{3}P$_1$–^{3}P$_2$	370.4151	2.65×10^{-7}	–	(3000)
C II	^{2}P$_{1/2}$–^{2}P$_{3/2}$	157.741	2.4×10^{-6}	1.80	47 (3000)
N II	^{3}P$_0$–^{3}P$_1$	205.3	2.07×10^{-6}	0.41	41
	^{3}P$_1$–^{3}P$_2$	121.889	7.46×10^{-6}	1.38	256
	^{3}P$_2$–^{1}D$_2$	0.65834	2.73×10^{-3}	2.99	7700
	^{3}P$_1$–^{1}D$_2$	0.65481	9.20×10^{-4}	2.99	7700
N III	^{2}P$_{1/2}$–^{2}P$_{3/2}$	57.317	4.8×10^{-5}	1.2	1880
O I	^{3}P$_2$–^{3}P$_1$	63.184	8.95×10^{-5}	–	2.3×10^4 (5×10^5)
	^{3}P$_1$–^{3}P$_0$	145.525	1.7×10^{-5}	–	3400 (1×10^5)
	^{3}P$_2$–^{1}D$_2$	0.63003	6.3×10^{-3}	–	1.8×10^6
O II	^{4}S$_{3/2}$–^{2}D$_{5/2}$	0.37288	3.6×10^{-5}	0.88	1160
	^{4}S$_{3/2}$–^{2}D$_{3/2}$	0.37260	1.8×10^{-4}	0.59	3890
O III	^{3}P$_0$–^{3}P$_1$	88.356	2.62×10^{-5}	0.39	461
	^{3}P$_1$–^{3}P$_2$	51.815	9.76×10^{-5}	0.95	3250
	^{3}P$_2$–^{1}D$_2$	0.50069	1.81×10^{-2}	2.50	6.4×10^5
	^{3}P$_1$–^{1}D$_2$	0.49589	6.21×10^{-3}	2.50	6.4×10^5
	^{1}D$_2$–^{1}S$_0$	0.43632	1.70	0.40	2.4×10^7
Ne II	^{2}P$_{1/2}$–^{2}P$_{3/2}$	12.8136	8.6×10^{-3}	0.37	5.9×10^5
Ne III	^{3}P$_2$–^{3}P$_1$	15.5551	3.1×10^{-2}	0.60	1.27×10^5
	^{3}P$_1$–^{3}P$_0$	36.0135	5.2×10^{-3}	0.21	1.82×10^4
Si II	^{2}P$_{1/2}$–^{2}P$_{3/2}$	34.8152	2.17×10^{-4}	7.7	(3.4×10^5)
S II	^{4}S$_{3/2}$–^{2}D$_{5/2}$	0.67164	2.60×10^{-4}	4.7	1240
	^{4}S$_{3/2}$–^{2}D$_{3/2}$	0.67308	8.82×10^{-4}	3.1	3270
S III	^{3}P$_0$–^{3}P$_1$	33.4810	4.72×10^{-4}	4.0	1780
	^{3}P$_1$–^{3}P$_2$	18.7130	2.07×10^{-3}	7.9	1.4×10^4
S IV	^{2}P$_{1/2}$–^{2}P$_{3/2}$	10.5105	7.1×10^{-3}	8.5	5.0×10^4
Ar II	^{2}P$_{1/2}$–^{2}P$_{3/2}$	6.9853	5.3×10^{-2}	2.9	1.72×10^6
Ar III	^{3}P$_2$–^{3}P$_1$	8.9914	3.08×10^{-2}	3.1	2.75×10^5
	^{3}P$_1$–^{3}P$_0$	21.8293	5.17×10^{-3}	1.3	3.0×10^4
Fe II	^{6}D$_{7/2}$–^{6}D$_{5/2}$	35.3491	1.57×10^{-3}	–	(3.3×10^6)
	^{6}D$_{9/2}$–^{6}D$_{7/2}$	25.9882	2.13×10^{-3}	–	(2.2×10^6)

Non-LTE

- Non-local thermodynamic equilibrium (NLTE or non-LTE) is a loose term which implies that the assumption of LTE fails.
  - Often, **statistical equilibrium** is implicitly assumed, usually with the **Maxwell distribution** and complete redistribution in frequency and angle. However, the populations are allowed to differ from the local Saha-Boltzmann equilibrium values.

- Statistical equilibrium
  - **[Rate equation]** Statistical equilibrium implies that the radiation fields (in all directions and on all frequencies) and level populations do not vary with time, as expressed in the statistical equilibrium equations:

$$\frac{dn_i(\vec{r})}{dt} = \sum_{j\neq i}^{N} n_j(\vec{r})P_{ji}(\vec{r}) - n_i(\vec{r})\sum_{j\neq i}^{N} P_{ij}(\vec{r}) = 0$$

Here, the transition rates for radiative bound-bound and collisional processes are given by

$$P_{ij} = R_{ij} + C_{ij} = A_{ij} + B_{ij}\overline{J}_{\nu 0} + C_{ij}$$

A similar expression holds for radiative bound-free rate but with $\overline{J}_{\nu 0}$ averaged over the ionization edge.

- **[Transport equations]** The population equations contain the mean intensities at all relevant frequencies via $\bar{J}_{\nu 0}$. The intensities are given by the radiative transfer equations:

$$\mu \frac{dI_\nu(\vec{r},\mu)}{d\tau_\nu(\vec{r})} = -S_\nu(\vec{r}) + I_\nu(\vec{r},\mu)$$

  at all frequencies ν, directions μ and locations $\vec{r}$.

  The rates P_{ij} depend on $\bar{J}_{\nu 0}$ and therefore on I_ν in other directions, whereas the optical depths τ_ν and the source functions S_ν depend on the populations n_l and n_u of the lower and upper levels.

  These populations may depend on other transitions and therefore on other populations again, each dependent on radiation fields at other frequencies.

  Thus, a given transition of interest may be influenced by many other transitions in the same particle species, or by transitions in other atoms and molecules if these poses transitions at overlapping frequencies.

  This intricate coupling between populations and radiation is non-linear and non-local. It can be very complex, except when the tremendous simplification of LTE may be assumed.

NLTE descriptions

- Departure coefficients
  - NLTE population departure coefficients b_i are defined as:

$$b_l = n_l / n_l^{\mathrm{LTE}} \qquad\qquad b_u = n_u / n_u^{\mathrm{LTE}}$$

  - Here, n is the actual population and n_{LTE} are the Saha-Boltzmann values for the lower and upper level.
- Formal temperatures
  - Another way to formalize the deviation of the source function from the Planck function is to introduce formal NLTE temperatures.
  - Excitation temperature, defined as the temperature to be entered into the Boltzmann distribution
  - Ionization temperature: the temperature that must be formally entered instead of LTE temperature in the Saha distribution.
  - Radiation temperature T_{rad} expresses the mean intensity into the Planck function by setting

$$B_\nu(T_{\mathrm{rad}}) \equiv J_\nu$$

  - Brightness temperature T_{b} expresses the observed intensity into the Planck function through

$$B_\nu(T_{\mathrm{b}}) \equiv I_\nu$$

Approximation Methods

Scattering Effects - (a) Pure Scattering

- Assumptions
  - ***isotropic scattering:*** scattered equally into equal solid angles
  - ***coherent scattering*** (elastic or monochromatic scattering): the total amount of radiation scattered per unit frequency is equal to the total amount absorbed in the same frequency range.
  - Thompson scattering (scattering from non-relativistic electrons) and dust scattering are nearly coherent.
- **scattering coefficient**

$$j_\nu^{\rm sca} = \alpha_\nu^{\rm sca} \int \Phi_\nu(\Omega' \to \Omega) I_\nu(\Omega') d\Omega$$

$$= \alpha_\nu^{\rm sca} \frac{1}{4\pi} \int I_\nu d\Omega' = \alpha_\nu^{\rm sca} J_\nu$$

In Rybicki & Lightman,
the scattering coefficient is denoted by σ_ν.

- **source function**

$$S_\nu = J_\nu = \frac{1}{4\pi} \int I_\nu d\Omega$$

- **radiative transfer equation**

$$\frac{dI_\nu}{ds} = -\alpha_\nu^{\rm sca} (I_\nu - J_\nu)$$

  - This is an integro-differential equation, and cannot be solved by the formal solution.

  $\rightarrow$ We need “Random walks, Rosseland approximation, or Eddington approximation.”

Random Walks in an "infinite" medium

- Random walks: let's consider a single photon rather than a beam of photons (i.e., ray).
  - <u>In an infinite, homogeneous medium</u>, net displacement of the photon after *N* free paths is zero, because the average displacement, being a vector, must be zero.

$$\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 + \cdots + \mathbf{r}_N \quad \rightarrow \quad \langle \mathbf{R} \rangle = 0$$

  - Root mean square net displacement:

$$\begin{aligned} l_*^2 \equiv \left\langle \mathbf{R^2} \right\rangle &= \left\langle \mathbf{r}_1^2 \right\rangle + \left\langle \mathbf{r}_2^2 \right\rangle + \left\langle \mathbf{r}_3^2 \right\rangle + \cdots + \left\langle \mathbf{r}_N^2 \right\rangle \\ &\quad + 2 \left\langle \mathbf{r}_1 \cdot \mathbf{r}_2 \right\rangle + 2 \left\langle \mathbf{r}_1 \cdot \mathbf{r}_3 \right\rangle + \cdots \\ &\approx N l^2 \quad \Leftarrow \quad \text{Note that } \left\langle \mathbf{r}_i^2 \right\rangle \approx l^2, \ \left\langle \mathbf{r}_i \cdot \mathbf{r}_j \right\rangle = 0 \ (i \neq j) \end{aligned}$$

$$\langle \mathbf{r}_i \cdot \mathbf{r}_j \rangle = \frac{1}{4\pi} \int_0^{2\pi} \int_{-1}^{1} \mu d\mu d\phi = 0$$

$$\boxed{\therefore \ l_* = \sqrt{N} l}$$

Here, the cross terms involve averaging the cosine of the angle between the directions before and after scattering, and this vanishes for isotropic scattering and for any scattering with front-back symmetry (Thomson or Rayleigh scattering).

Random Walks in "finite" medium

- In a finite medium, a photon generated somewhere within the medium will scatter until it escapes completely.
  - For regions of large optical depth, the mean number of scatterings to escape is roughly determined by $l_* \approx L$ (net displacement = the typical size of the medium).

$$l_* = \sqrt{N} l \approx L \qquad \Rightarrow \quad N \approx L^2/l^2 = L^2 (n\sigma_\nu^{\rm sca})^2$$
$$\approx \tau^2 \quad (\tau \gg 1)$$

$$N \approx \frac{1}{2}\tau_0^2 \quad \text{in a sphere}$$
$$\approx \frac{3}{2}\tau_0^2 \quad \text{in an infinite slab}$$

(Seon et al. 2023, JKAS)

  - For regions of small optical depth, the probability of scatterings within τ is

$$\int_0^\tau P(t)dt = 1 - e^{-\tau} \approx \tau \qquad \therefore \ N \approx \tau \quad (\tau \ll 1)$$

  - In summary, for any optical thickness, the mean number of scatterings is

$$N \approx \tau^2 + \tau \quad \text{or} \quad N \approx \max(\tau, \tau^2)$$

  - In the case of Ly-alpha RT, $N \approx \tau$ (see Seon & Kim 2020, ApJS).

Scattering Effects - (b) Combined Scattering and Absorption

- The RT equation including the scattering and absorption:

$$\boxed{\frac{dI_\nu}{ds} = -\alpha_\nu^{\mathrm{ext}} I_\nu + j_\nu + \alpha_\nu^{\mathrm{scatt}} \int \Phi_\nu(\Omega' \rightarrow \ \Omega) I_\nu(\Omega') d\Omega'}$$

- Assuming ***the LTE condition and an isotropic scattering***, we obtain

$$j_\nu = \alpha_\nu^{\mathrm{abs}} B_\nu$$

$$\begin{aligned} j_\nu^{\mathrm{sca}} &= \alpha_\nu^{\mathrm{sca}} \int \Phi_\nu(\Omega' \rightarrow \Omega) I_\nu(\Omega') d\Omega \\ &= \alpha_\nu^{\mathrm{sca}} J_\nu \end{aligned} \qquad \Rightarrow \boxed{\frac{j_\nu^{\mathrm{sca}}}{\alpha_\nu^{\mathrm{sca}}} = J_\nu}$$

This indicates that ***the mean intensity plays a role of source function for scattering***.

- Then, the RT equation becomes

$$\boxed{\frac{dI_\nu}{ds} = -(\alpha_\nu^{\mathrm{abs}} + \alpha_\nu^{\mathrm{sca}}) I_\nu + \alpha_\nu^{\mathrm{abs}} B_\nu + \alpha_\nu^{\mathrm{sca}} J_\nu}$$

- Therefore, the RT equation in the case of combined absorption and scattering can be written:

$$
\begin{aligned}
\frac{dI_\nu}{ds} &= -\alpha_\nu^{\rm abs}\left(I_\nu - B_\nu\right) - \alpha_\nu^{\rm sca}\left(I_\nu - J_\nu\right) \\
&= -\left(\alpha_\nu^{\rm abs} + \alpha_\nu^{\rm sca}\right)\left(I_\nu - S_\nu\right) \\
&= -\alpha_\nu^{\rm ext}\left(I_\nu - S_\nu\right)
\end{aligned}
$$

$$
\Leftarrow \ \text{Here, } S_\nu \equiv \frac{\alpha_\nu^{\rm abs} B_\nu + \alpha_\nu^{\rm sca} J_\nu}{\alpha_\nu^{\rm abs} + \alpha_\nu^{\rm sca}}
$$

$$
\alpha_\nu^{\rm ext} \equiv \alpha_\nu^{\rm abs} + \alpha_\nu^{\rm sca}
$$

- ***The source function is a weighted average of the two source functions (for absorption and scattering).***

- Extinction coefficient: $\alpha_\nu^{\rm ext} \equiv \alpha_\nu^{\rm abs} + \alpha_\nu^{\rm sca}$
- Optical depth: $d\tau_\nu \equiv \alpha_\nu^{\rm ext} ds$

- If a matter element is deep inside a medium (i.e., in the thermodynamic equilibrium),

$$
J_\nu = B_\nu \quad \rightarrow \quad S_\nu = B_\nu
$$

- If the element is isolated in free space, where $J_\nu = 0$, the the source function becomes

$$
S_\nu = (\alpha_\nu^{\rm abs}/\alpha_\nu^{\rm ext}) B_\nu
$$

- Generalized mean free path:

$$l_\nu = \left(\alpha_\nu^{\rm abs} + \alpha_\nu^{\rm scat}\right)^{-1}$$

- Probability of a (random walk) step ending in absorption:

$$\epsilon_\nu = \alpha_\nu^{\rm abs}/\alpha_\nu^{\rm ext}$$

- Probability for scattering (known as the ***single-scattering albedo*** at frequency ν)

$$a_\nu = 1 - \epsilon_\nu = \alpha_\nu^{\rm sca}/\alpha_\nu^{\rm ext}$$

- Source function:

$$\begin{aligned} S_\nu &= \epsilon_\nu B_\nu + (1-\epsilon_\nu) J_\nu \\ &= (1-a_\nu) B_\nu + a_\nu J_\nu \end{aligned}$$

Random Walks with Scattering and Absorption

- ***<u>In an infinite medium</u>***, every photon is eventually absorbed.
  - Since a random walk can be terminated with probability $\epsilon = \alpha^{\rm abs}/\alpha^{\rm ext}$ at the end of each free path, the mean number of free paths is given by

mean number of free paths × probability of termination = 1

$$N\epsilon = 1 \quad \Rightarrow \quad N = 1/\epsilon$$

  - Diffusion length (thermalization length, effective mean path, or effective free path): a measure of the net displacement between the points of creation and destruction of a typical photon.

$$\begin{aligned} l_* &\approx \sqrt{N} l = l/\sqrt{\epsilon} \\ &\approx (\alpha_\nu^{\rm ext})^{-1}\sqrt{\alpha_\nu^{\rm ext}/\alpha_\nu^{\rm abs}} \\ &\approx \left(\alpha_\nu^{\rm abs}\alpha_\nu^{\rm ext}\right)^{-1/2} \end{aligned}$$

- ***In a finite medium***:
  - The behavior depends on whether its size L is larger or smaller than the effective mean path l_*.
  - Effective optical thickness: $\tau_* = L/l_* \approx \sqrt{\tau_{\rm abs}\left(\tau_{\rm abs} + \tau_{\rm sca}\right)} = \sqrt{\tau_{\rm abs}\tau_{\rm ext}}$

    $$\text{where} \;\; \tau_{\rm abs} \equiv \alpha_\nu^{\rm abs} L, \;\; \tau_{\rm sca} \equiv \alpha_\nu^{\rm scatt} L, \;\; \tau_{\rm ext} \equiv \alpha_\nu^{\rm ext} L$$

  - If effectively thin or translucent ($\tau_* \ll 1,\ L \ll l_*$), most photons will escape the medium before being destroyed.
    - Luminosity of thermal source with volume V is

$$\boxed{L_\nu = 4\pi j_\nu V = 4\pi\alpha_\nu^{\rm abs} B_\nu V \;\; (\tau_* \ll 1)}$$

  - If effectively thick ($\tau_* \gg 1,\ L \gg l_*$), we expect $I_\nu \to B_\nu$ and $S_\nu \to B_\nu$, and only the photons emitted within an effective path length of the boundary will have a reasonable chance of escaping before being absorbed.

$$\boxed{L_\nu = 4\pi\alpha_\nu^{\rm abs} B_\nu A l_* = 4\pi\sqrt{\epsilon_\nu} B_\nu A \;\; (\tau_* \gg 1)} \quad \Leftarrow \quad \begin{array}{l} V \approx A l_* \\ l_* \approx \left(\alpha_\nu^{\rm abs}\alpha_\nu^{\rm ext}\right)^{-1/2} \end{array}$$

l_* is sometimes called the thermalization length, since it describes the distance over which TE of the radiation is established.

Approximate Solutions

- How to solve the radiative transfer equation:

$$\frac{dI_\nu}{ds} = -\alpha_\nu^{\mathrm{ext}} \left(I_\nu - S_\nu\right)$$

$$S_\nu = \left(1-\epsilon_\nu\right) J_\nu + \epsilon_\nu B_\nu \quad \text{and} \quad \epsilon_\nu = \alpha_\nu^{\mathrm{abs}} / \alpha_\nu^{\mathrm{ext}}$$

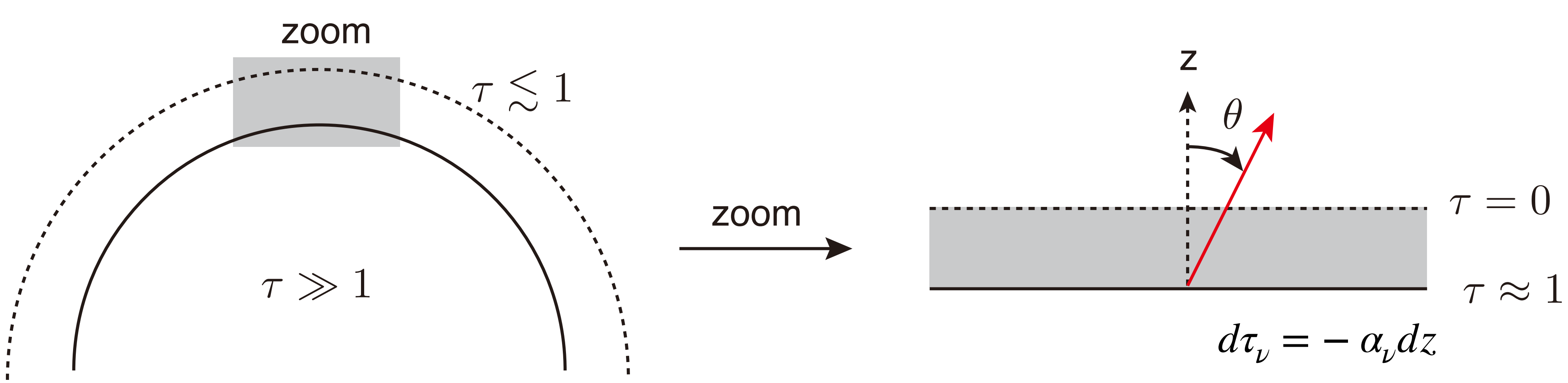
under the assumption of the isotropic scattering

- We will discuss three approximations to solve the above equation in a plane-parallel medium, primarily developed for the stellar atmosphere.
  - Rosseland approximation
  - Eddington approximation
  - Grey atmosphere

- Additionally, we discuss an iteration method to solve the scattering RT.
  - Iteration method

Plane-parallel Atmosphere

- Stellar Atmosphere
  - The atmosphere of a star is defined as its outer regions that determine the properties of the radiative flux that emanates from its surface.
  - The calculations of the structure of a stellar atmosphere necessitate the knowledge of the detailed monochromatic opacity and can only be undertaken with powerful numerical capabilities such as those provided by modern computers.
  - However, by making appropriate approximations, certain interesting results can be obtained analytically.
- ***Plane-parallel atmosphere***
  - We ignore the curvature of the underlying body. In a plane-parallel medium with a frequency independent opacity. This simplification is especially suitable for models of bodies where the atmosphere is a relatively thin layer around the body, i.e., "locally virtually flat."
  - The optical depth is defined as a function of the vertical distance *z* rather than the distance long the ray path *s*. It increases as we go deeper down. The normal convention is to take $\tau = 0$ at the top of the stellar atmosphere

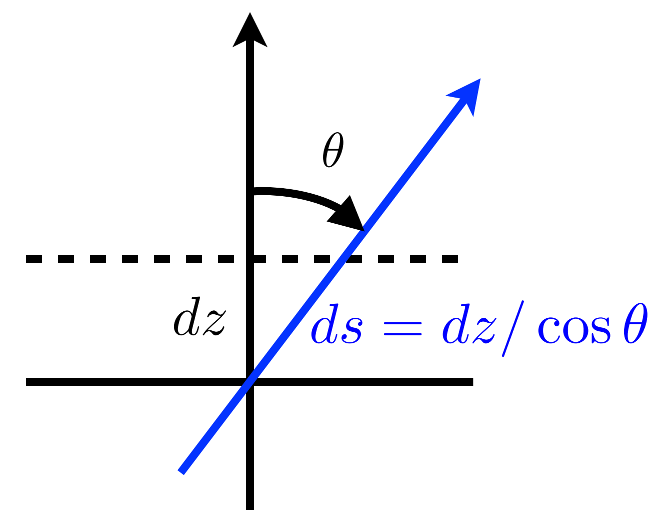


(1) Radiative Diffusion: Rosseland Approximation

- Imagine a **plane-parallel medium** (in which $\rho,\ T$ depend only on depth *z*).

$$ds = \frac{dz}{\cos\theta} = \frac{dz}{\mu} \quad \Rightarrow \quad \mu \frac{\partial I_\nu(z,\mu)}{\partial z} = -\alpha_\nu^{\rm ext}\left(I_\nu - S_\nu\right)$$

$$\Rightarrow \boxed{I_\nu(z,\mu) = S_\nu - \frac{\mu}{\alpha_\nu^{\rm ext}} \frac{\partial I_\nu}{\partial z}}$$



- "zeroth" order approximation: when the point in question is deep in the material, all quantities changes slowly on the scale of a mean free path $\ell = 1/\alpha^{\rm ext}$ and the above derivative term is very small.

$$I_\nu^{(0)}(z,\mu) \approx S_\nu^{(0)}(T) \qquad \therefore J_\nu^{(0)} = S_\nu^{(0)} \ \ \text{and} \ \ I_\nu^{(0)} = S_\nu^{(0)} = B_\nu$$

LTE condition

This is independent of the angle.

- "first" order approximation:

$$\boxed{\begin{aligned} I_\nu^{(1)}(z,\mu) &\approx S_\nu^{(0)} - \frac{\mu}{\alpha_\nu^{\rm ext}} \frac{\partial I_\nu^{(0)}}{\partial z} \\ &= S_\nu^{(0)} - \frac{\mu}{\alpha_\nu^{\rm ext}} \frac{\partial B_\nu(T)}{\partial z} \end{aligned}}$$

$\Leftarrow$ linear in μ

- Net specific flux along z : the angle-independent part of the intensity does not contribute to the flux.

$$F_\nu(z) = \int I_\nu^{(1)}(z,\mu)\cos\theta d\Omega = 2\pi\int_{-1}^{+1} I_\nu^{(1)}(z,\mu)\mu d\mu$$
$$= -\frac{2\pi}{\alpha_\nu^{\rm ext}}\frac{\partial B_\nu}{\partial z}\int_{-1}^{+1}\mu^2 d\mu$$
$$= -\frac{4\pi}{3\alpha_\nu^{\rm ext}}\frac{\partial B_\nu(T)}{\partial T}\frac{\partial T}{\partial z}$$

- Total integrated flux:

$$F(z) = \int_0^\infty F_\nu(z)d\nu = -\frac{4\pi}{3}\frac{\partial T}{\partial z}\int_0^\infty \frac{1}{\alpha_\nu^{\rm ext}}\frac{\partial B_\nu}{\partial T}d\nu$$

Let’s define the **Rosseland mean absorption coefficient**

$$\boxed{\frac{1}{\alpha_{\rm R}} = \frac{\displaystyle\int_0^\infty \frac{1}{\alpha_{\rm ext}}\frac{\partial B_\nu}{\partial T}d\nu}{\displaystyle\int_0^\infty \frac{\partial B_\nu}{\partial T}d\nu}}$$

Use

$$\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu = \frac{\partial}{\partial T}\int_0^\infty B_\nu d\nu = \frac{\partial\left(\sigma_{\rm SB}T^4/\pi\right)}{\partial T} = \frac{4\sigma_{\rm SB}T^3}{\pi}$$

Then, we obtain the Rosseland approximation to radiative flux.

$$\boxed{F(z) = -\frac{16\sigma_{\rm SB}T^3}{3\alpha_{\rm R}}\frac{\partial T}{\partial z} \quad \Rightarrow \quad F(z) = -\chi\nabla T}$$

which is also called ***the equation of radiative diffusion***. This equation is one of the fundamental equations for studying stellar interiors.

- The flux equation can be interpreted as a heat conduction with an **"effective heat conductivity."**

$$\boxed{\chi = \frac{16\sigma_{\rm SB}T^3}{3\alpha_{\rm R}}}$$

- At which frequencies the Rosseland mean becomes important?
    - The mean involves a weighted average of $1/\alpha_\nu^{\rm ext}$ so that frequencies at which **the extinction coefficient is small (transparent) tend to dominate.**
    - The weighting function $\partial B_\nu/\partial T$ has a shape similar to that of the Planck function, but it peaks at $h\nu_{\rm max} = 3.8k_{\rm B}T$, instead of $h\nu_{\rm max} = 2.8k_{\rm B}T$.

(2) Eddington Approximation

- In Eddington approximation, the intensities are assumed to approach isotropy, and not necessarily their thermal values.

  In the Rosseland approximation, the intensities approach the Planck function at large effective depths.

- Near isotropy can be introduced by assuming that the intensity is linear in μ. (frequency is suppressed for convenience).

$$I(\tau,\mu) = a(\tau) + b(\tau)\mu$$

- Let us take the first three moments.

  mean intensity: $$J \equiv \frac{1}{2}\int_{-1}^{+1} I d\mu = a$$

  flux: $$H \equiv \frac{1}{2}\int_{-1}^{+1} \mu I d\mu = \frac{b}{3} \quad \rightarrow \quad \boxed{K = \frac{1}{3}J}$$

  radiation pressure: $$K \equiv \frac{1}{2}\int_{-1}^{+1} \mu^2 I d\mu = \frac{a}{3}$$

  This relation is called the **Eddington approximation**

  - Compare with the following equations for the isotropic radiation.

$$p = \frac{1}{3}u \quad \left(p \equiv \frac{1}{c}\int I \cos^2\theta d\Omega,\ u(\mathbf{\Omega}) = \frac{1}{c}I \right)$$

- optical depth and the transfer equation:

$$d\tau(z) \equiv -\alpha^{\text{ext}} dz, \quad \mu \frac{\partial I}{\partial \tau} = I - S$$

Note: source function is independent to μ (because $S = (1-\epsilon)J + \epsilon B$).

- Integrate the above equation and obtain the following equations.

$$\frac{1}{2}\int_{-1}^{+1} d\mu \left(\mu \frac{\partial I}{\partial \tau} = I - S \right) \rightarrow \frac{\partial H}{\partial \tau} = J - S$$

$$\frac{1}{2}\int_{-1}^{+1} d\mu \mu \left(\mu \frac{\partial I}{\partial \tau} = I - S \right) \rightarrow \frac{\partial K}{\partial \tau} = H \rightarrow \frac{1}{3}\frac{\partial J}{\partial \tau} = H$$

- The two equations can be combined to yield:

$$\frac{1}{3}\frac{\partial^2 J}{\partial \tau^2} = J - S \quad \rightarrow \boxed{\frac{1}{3}\frac{\partial^2 J}{\partial \tau^2} = \epsilon(J - B)} \longleftarrow S = (1-\epsilon)J + \epsilon B$$

Let us define a new optical depth $\tau_* \equiv \sqrt{3\epsilon}\tau = \sqrt{3\tau_{\text{abs}}(\tau_{\text{abs}} + \tau_{\text{sca}})}$

The radiative equation is then $\boxed{\frac{\partial^2 J}{\partial \tau_*^2} = J - B}$

- This equation is sometimes called the ***radiative diffusion equation***. Given the temperature structure of the medium, $B(\tau)$, the equation can be solved for J.

- Boundary conditions: Two-stream approximation

  - To solve the second order differential equation, we need two boundary conditions. The boundary conditions can be provided in several ways. One way to do is to use two-stream approximation, in which the entire radiation field is represented by radiation at just two angles, i.e., $\mu = \pm\,\mu_0$:

$$I(\tau,\mu) = I^+(\tau)\delta(\mu-\mu_0) + I^-(\tau)\delta(\mu+\mu_0)$$

  - The two terms denote the outward and inward intensities. Then, the three moments are

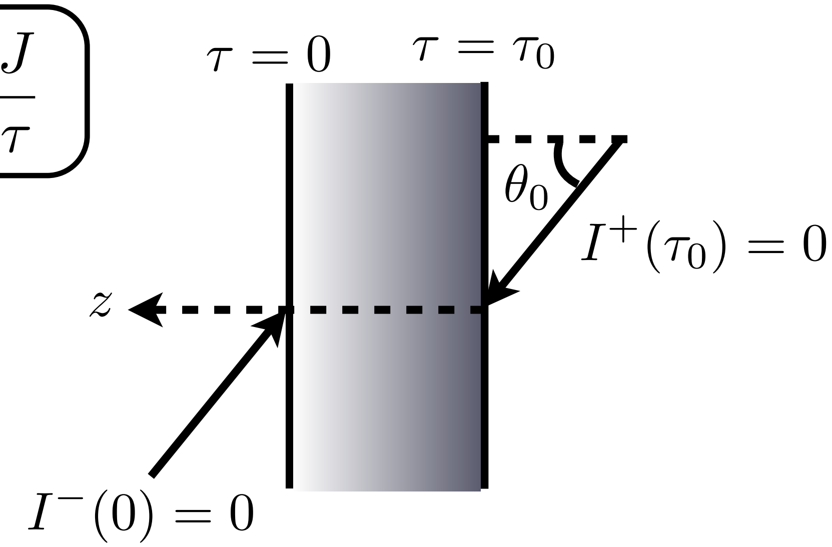
$$J = \frac{1}{2}\left(I^+ + I^-\right)$$
$$H = \frac{1}{2}\mu_0\left(I^+ - I^-\right)$$
$$K = \frac{1}{2}\mu_0^2\left(I^+ + I^-\right)$$

$$\rightarrow \text{ we obtain } \mu_0 = \frac{1}{\sqrt{3}} \text{ in order to satisty } K = \frac{1}{3}J$$

$$(\theta_0 = \cos^{-1}\mu_0 = 54.74^\circ)$$

  - Using $H = \dfrac{1}{3}\dfrac{\partial J}{\partial \tau}$, we obtain: $\boxed{I^+ = J + \dfrac{1}{3}\dfrac{\partial J}{\partial \tau},\ \ I^- = J - \dfrac{1}{3}\dfrac{\partial J}{\partial \tau}}$

As an example, suppose the medium extends from $\tau = 0$ to $\tau = \tau_0$ and there is no incident radiation (from both sides). Then, we obtain two boundary conditions:



$$I^+(\tau_0) = 0 \ \text{ and } \ I^-(0) = 0 \ \rightarrow \ \boxed{\begin{array}{ll} \dfrac{1}{\sqrt{3}}\dfrac{\partial J}{\partial \tau} = J & \text{at } \ \tau = 0 \\ \dfrac{1}{\sqrt{3}}\dfrac{\partial J}{\partial \tau} = -J & \text{at } \ \tau = \tau_0 \end{array}}$$

These two conditions are sufficient to determine the solution.

(3) Grey Atmosphere

- Grey Atmosphere
  - For example, by supposing that **the opacity is independent of frequency**, some illustrative and insightful results can be found. A stellar atmosphere while assuming such an opacity spectrum is called a grey atmosphere.
  - Nevertheless, the radiation field present in such an atmosphere will posses a frequency distribution since matter emits a radiative spectrum associated to its temperature.
  - We will find an expression for the temperature profile (with respect to depth) and the flux emanating from a grey atmosphere.
  - It is more convenient to use a convention where τ increases inward. Then, the RT equation is given by

$$\boxed{\mu \frac{dI_\nu}{d\tau} = I_\nu - S_\nu}$$

Here, τ is independent of frequency.

  - (1) By integrating this equation over all frequencies, the RT equation becomes

$$\boxed{\mu \frac{dI}{d\tau} = I - S}$$

where $I = \int_0^\infty I_\nu d\nu, \quad S = \int_0^\infty S_\nu d\nu$

- **Radiative equilibrium**
  - If there exists no energy sources or sinks in a stellar atmosphere, the amount of energy absorbed by a given layer is equal to the energy that it emits.
  - In the plane-parallel approximation, this assumption implies that the energy generated in the stellar interior passes out in the form of a constant energy flux through the outer layers of the atmosphere since the energy emitted is distributed over the same area independently of depth.

$$\int\int \alpha_\nu I_\nu d\nu d\Omega = \int\int j_\nu d\nu d\Omega$$

total energy absorbed (or scattered by matter) = energy that it radiates

  - In the **grey atmosphere**, using the definition of the source function, we obtain

$$\int\int \alpha_\nu I_\nu d\nu d\Omega = \int\int \alpha_\nu S_\nu d\nu d\Omega \quad \rightarrow \quad \alpha \int J_\nu d\nu = \alpha \int S_\nu d\nu$$

$$\boxed{J = S}$$

- (2) Average the first equation over the solid angle:

$$\frac{1}{2}\int_{-1}^{1} d\mu \left(\mu \frac{dI_\nu}{d\tau}\right) = \frac{1}{2}\int_{-1}^{1} d\mu I_\nu - \frac{1}{2}\int_{-1}^{1} d\mu S_\nu$$

$$\boxed{\frac{dH}{d\tau} = J - S \quad \rightarrow \quad \frac{dH}{d\tau} = 0 \quad \text{from the radiative equilibrium } (J = S) \qquad H = \mathrm{constant}}$$

- (3) Multiply the first equation by μ and average over the solid angle:

$$\frac{1}{2}\int_{-1}^{1} d\mu \left(\mu^2 \frac{dI}{d\tau}\right) = \frac{1}{2}\int_{-1}^{1} d\mu \,(\mu I) - \frac{1}{2}\int_{-1}^{1} d\mu \,(\mu S)$$

$$\boxed{\frac{dK}{d\tau} = H}$$

if the source function is isotropic.

- Integrating over the above equation, we obtain the pressure as a function optical depth:

$$\boxed{K(\tau) = H \times (\tau + c)}$$

Here, Hc is the constant of integration.

- ***Using the Eddington approximation***, the mean intensity is obtained to be

$$K = \frac{1}{3}J \longrightarrow \boxed{J(\tau) = 3H \times (\tau + c)}$$

$c = \dfrac{2}{3}$ can be obtained as shown in the next slide

Derivation of the constant of integration:

$$\mu \frac{dI}{d\tau} = I - S \Rightarrow$$

$\mu > 0$

$$e^{-\tau/\mu}\frac{dI}{d(\tau/\mu)} - e^{-\tau/\mu} I = -e^{-\tau/\mu} S$$

$$\frac{d}{d(\tau/\mu)}\left(e^{-\tau/\mu} I\right) = -e^{-\tau/\mu} S$$

$$\left[e^{-\tau'/\mu} I\right]_{\tau'=\infty}^{\tau} = -\int_{\infty}^{\tau} e^{-\tau'/\mu} S d(\tau'/\mu)$$

$$I(\tau,\mu) = \int_{\tau}^{\infty} e^{-(\tau'-\tau)/\mu} S d(\tau'/\mu)$$

$\mu < 0$

$$e^{\tau/|\mu|}\frac{dI}{d(\tau/|\mu|)} + e^{\tau/|\mu|} I = e^{\tau/|\mu|} S$$

$$\frac{d}{d(\tau/|\mu|)}\left(e^{\tau/|\mu|} I\right) = e^{\tau/|\mu|} S$$

$$\left[e^{\tau'/|\mu|} I\right]_{\tau'=0}^{\tau} = \int_{0}^{\tau} e^{\tau'/|\mu|} S d(\tau'/|\mu|)$$

$$I(\tau,\mu) = \int_{0}^{\tau} e^{-(\tau-\tau')/\mu} S d(\tau'/|\mu|)$$

$$\mu > 0: \;\; I(0,\mu) = \int_0^{\infty} e^{-\tau'/\mu} S d(\tau'/\mu) = 3H \int_0^{\infty} e^{-\tau'/\mu} (\tau' + c) d(\tau'/\mu) \longrightarrow \boxed{I(0,\mu) = 3H(\mu + c)}$$

Now, from the definition of the outward flux H at $\tau = 0$:

$$H = \frac{1}{2}\int_{-1}^{1} \mu I(0,\mu) d\mu = \frac{1}{2}\int_{0}^{1} \mu I(0,\mu) d\mu = \frac{3H}{2}\left(\frac{1}{3} + \frac{c}{2}\right)$$

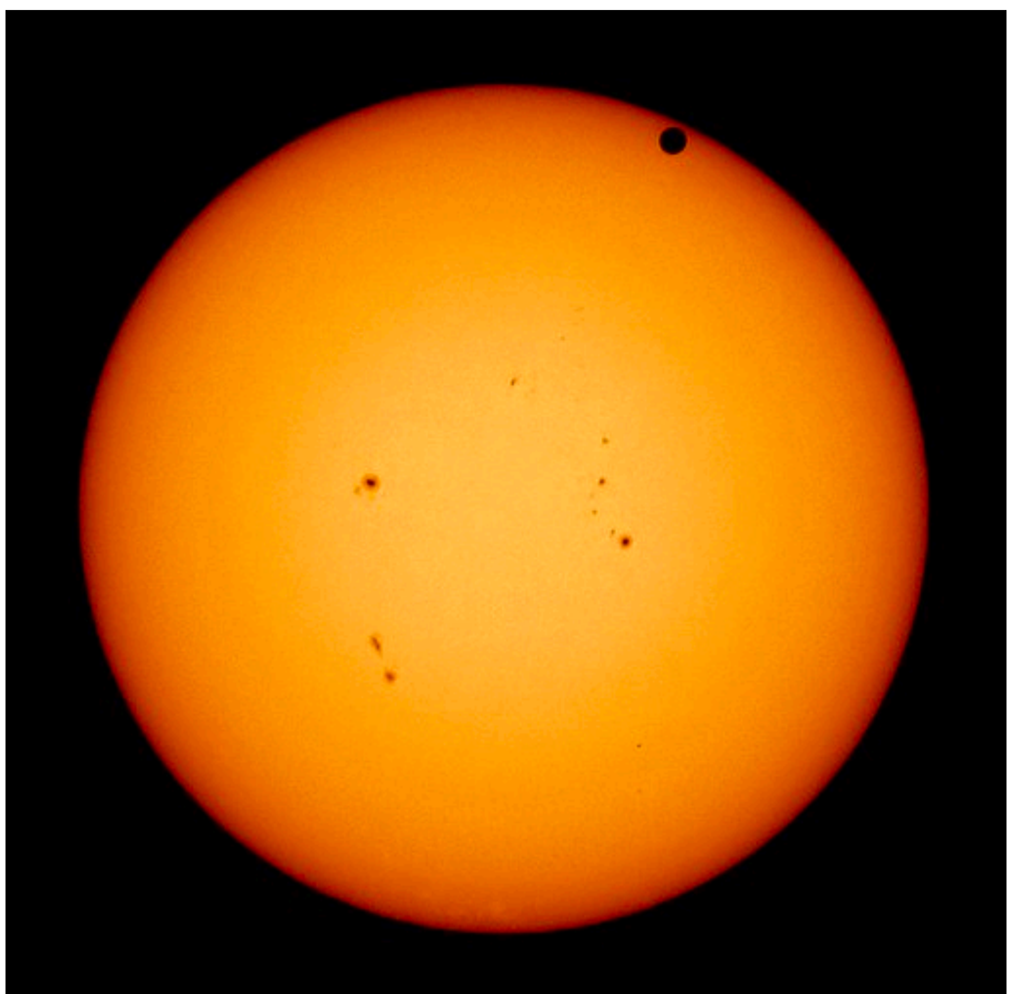
Note that $I(0,\mu) = 0$ for $\mu < 0$.

This gives the value of the constant of integration to be

$$\boxed{c = \frac{2}{3}}$$

Limb Darkening

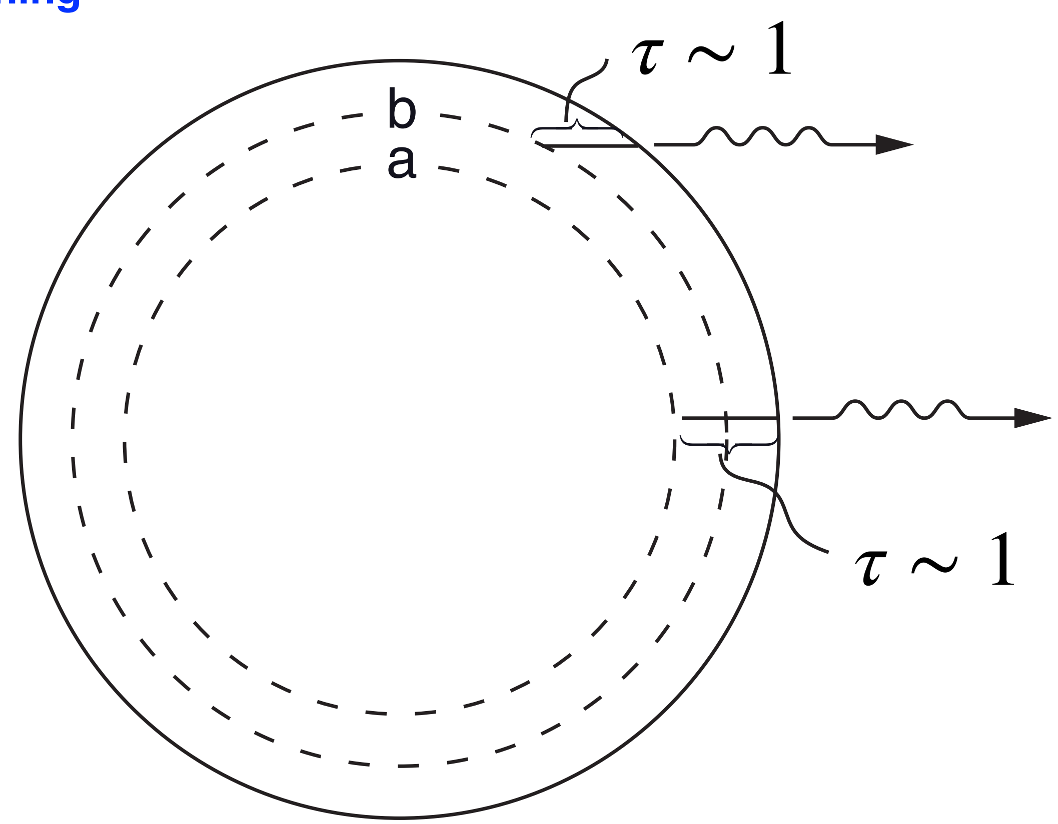
- Observations show that the radiation intensity at the center of the solar disk is larger than the corresponding intensity detected near its limb. This is called the limb darkening effect.
- Since the local temperature in the atmosphere increases with depth and the radiation leaving the star comes from layers with optical depth approximately equal to ~ 1, **the layers responsible for this radiation are hotter in the center than near the limb.**
- The Eddington limb darkening law can be estimated with the results found for the grey atmosphere.



[credit] Brocken Inaglory

Eddington limb darkening

$$\frac{I_\nu(\mu)}{I_\nu(1)} = \frac{3}{5}\left(\mu + \frac{2}{3}\right)$$



$T_a > T_b \Rightarrow S_a > S_b$

- Limb Darkening
  - The intensity at the surface of star is

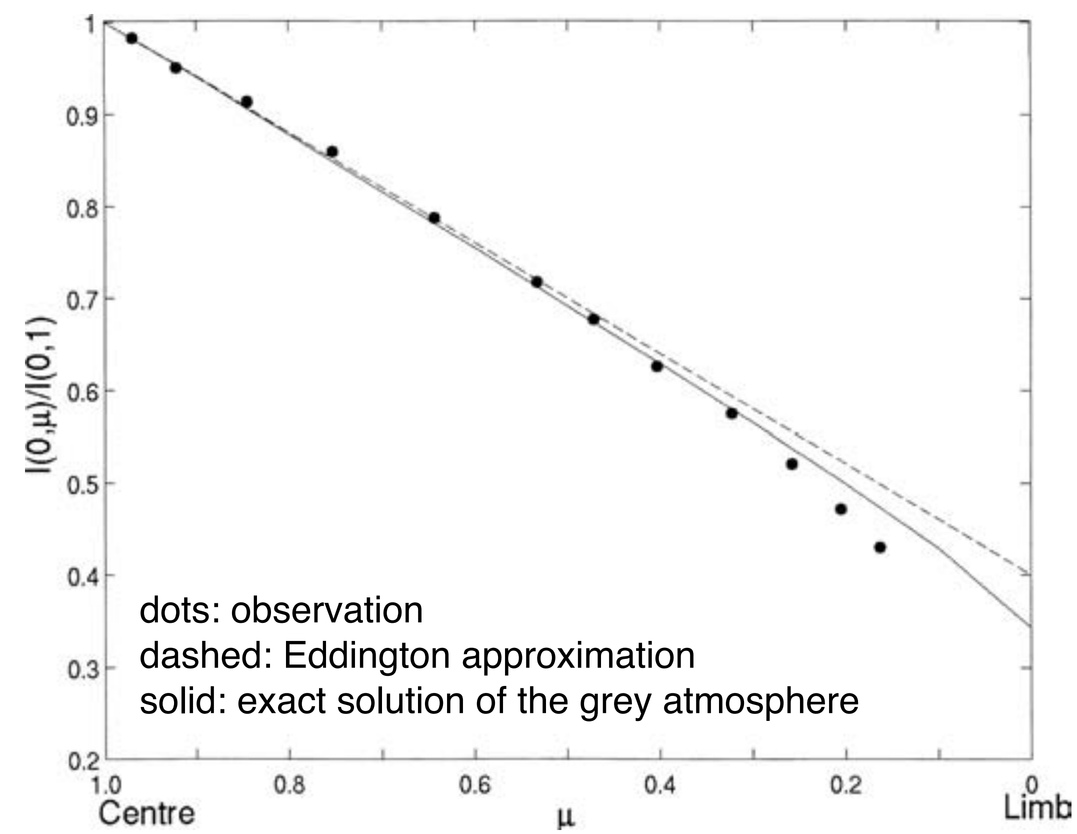
$$I(\tau=0,\mu) = \int_0^\infty S(\tau) e^{-\tau/\mu} \frac{d\tau}{\mu}$$
$$= \int_0^\infty 3H\left(\tau + \frac{2}{3}\right) e^{-\tau/\mu} d(\tau/\mu)$$
$$= 3H\left(\mu + \frac{2}{3}\right)$$

$$S = J = 3H\left(\tau + \frac{2}{3}\right)$$

  - The direction of the intensity coming from the center of the disk defines the angle $\theta = 0$ or $\mu = 1$, while the intensity near the limb of the disk is related to values of $\mu < 1$. The ratio of these intensities is

$$\boxed{\frac{I(0,\mu)}{I(0,1)} = \frac{3}{5}\left(\mu + \frac{2}{3}\right)}$$

This is called the Eddington limb darkening law.



for wavelength $\lambda = 5485\text{Å}$ [Pierce et al. 1950]

  - At the limb ($\mu = 0$) this ratio gives 0.4. Therefore, this demonstrates that the intensity for the grey-atmosphere approximation at the limb is 40% of the value of that at the stellar disk's center.

- Temperature Profile
  - From the definition of the effective temperature:

$$H = \frac{1}{4\pi}\int \cos\theta I d\Omega = \frac{1}{4\pi}F = \frac{1}{4\pi}\sigma_{\rm SB}T_{\rm eff}^4$$

  - Relation between the intensity and flux that comes outward:

$$F = \pi J \longrightarrow \quad J(\tau) = \frac{1}{\pi}F(\tau) = \frac{1}{\pi}\sigma_{\rm SB}T^4$$

Here, T is the effective temperature at τ.

  - Combining these, we obtain

$$J(\tau) = 3H\left(\tau + \frac{2}{3}\right) \longrightarrow \quad T^4 = \frac{3}{4}T_{\rm eff}^4\left(\tau + \frac{2}{3}\right)$$

$$\boxed{T = T_{\rm eff}\left[\frac{3}{4}\left(\tau + \frac{2}{3}\right)\right]^{1/4}}$$

  - This equation tells us how temperature varies inside a grey atmosphere.

(4) Iteration Method

- Recall
$$\frac{dI(s)}{ds} = -\alpha^{\rm ext} I(s) + \alpha^{\rm sca} \int \Phi(\Omega, \Omega') I(s, \Omega') d\Omega' + j(s)$$
or
$$\frac{dI(\tau)}{d\tau} = -I(\tau) + a \int \Phi(\Omega, \Omega') I(\tau, \Omega') d\Omega' + \mathsf{S}(\tau) \quad \left(d\tau \equiv \alpha^{\rm ext} ds, \ \mathsf{S}(\tau) \equiv \frac{j(\tau)}{\alpha^{\rm ext}} \right)$$

- Let I_0 be the intensity of photons that come directly from the source, I_1 the intensity of photons that have been scattered once by dust, and I_n the intensity after *n* scatterings. Then,
$$I(s) = \sum_{n=0}^{\infty} I_n(s)$$

- The intensities I_n satisfy the equations.
$$\begin{aligned} \frac{dI_0(\tau)}{d\tau} &= -I_0(\tau) + \mathsf{S}(\tau) \\ \frac{dI_n(\tau)}{d\tau} &= -I_n(\tau) + a \int \Phi(\Omega, \Omega') I_{n-1}(\tau, \Omega') d\Omega' \\ &\equiv -I_n(\tau) + \mathsf{S}_{n-1}(\tau) \qquad \left(\mathsf{S}_{n-1}(\tau) \equiv a \int \Phi(\Omega, \Omega') I_{n-1}(\tau, \Omega') d\Omega' \right) \end{aligned}$$

- Then, the formal solutions are:
$$\rightarrow \quad \begin{aligned} I_0(\tau) &= e^{-\tau} I_0(0) + \int_0^{\tau} e^{-(\tau-\tau')} \mathsf{S}(\tau') d\tau' \\ I_n(\tau) &= e^{-\tau} I_n(0) + \int_0^{\tau} e^{-(\tau-\tau')} \mathsf{S}_{n-1}(\tau') d\tau' \end{aligned}$$

(a) application to the edge-on galaxies

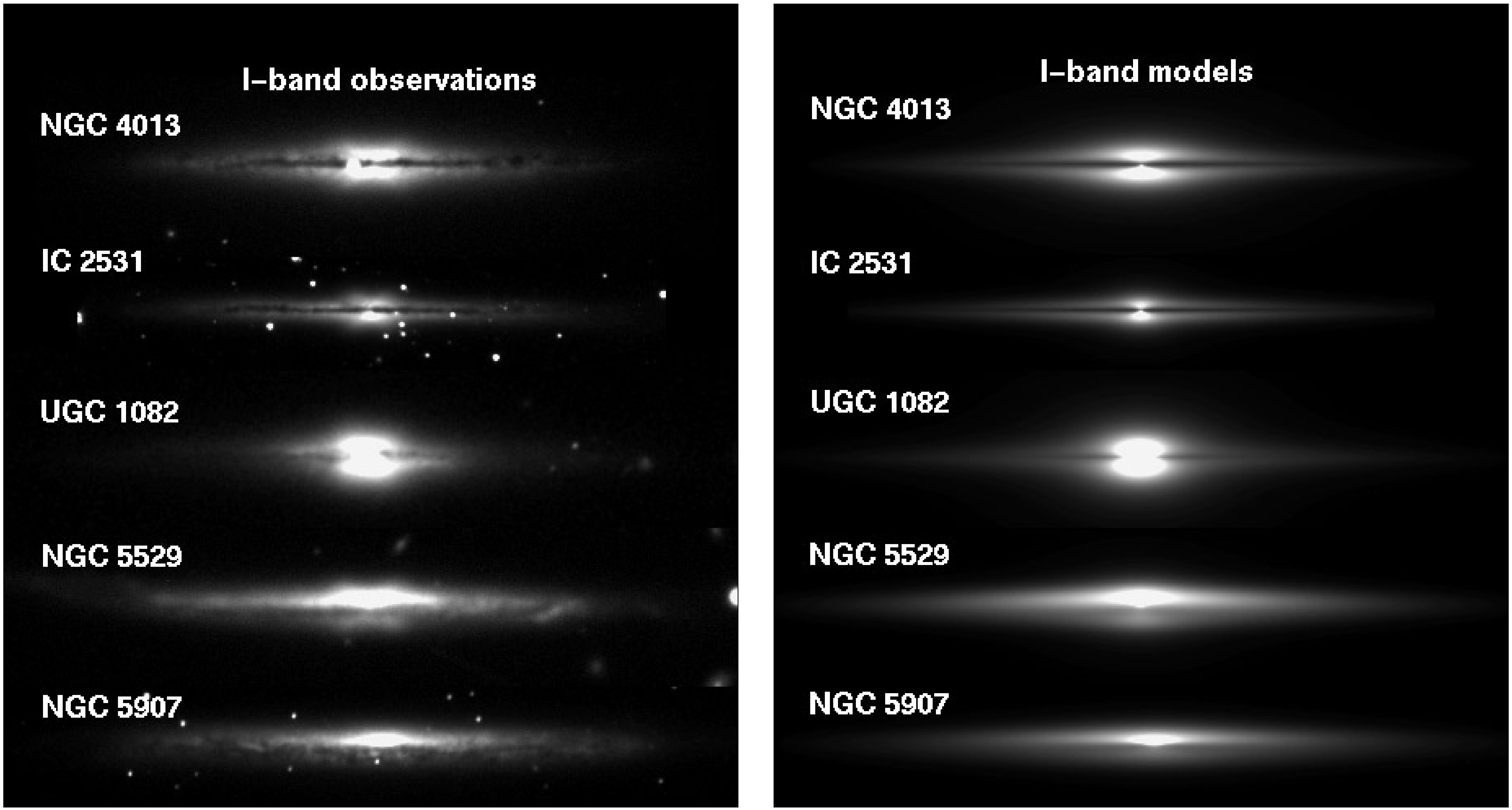
- The solution can be further simplified by assuming that

$$\frac{I_n}{I_{n-1}} \approx \frac{I_1}{I_0} \quad (n \geq 2)$$

- Then, the infinite series becomes

$$I \approx I_0 \sum_{n=0}^{\infty} \left(\frac{I_1}{I_0} \right)^n = \frac{I_0}{1 - I_1/I_0}$$

- Kylafis & Bahcall (1987) and Xilouris et al. (1997, 1998, 1999) applied this approximation to model the dust radiative transfer process in the edge-on galaxies.



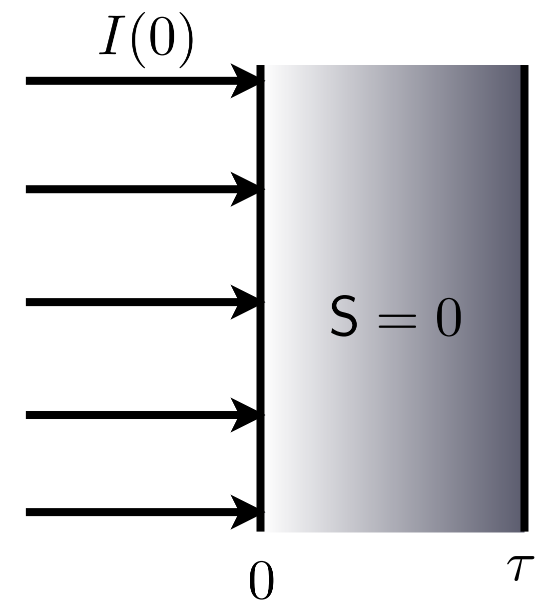
Xilouris et al. (1999)

(b) solution for the perfect forward scattering

- Assume the very strong forward-scattering

$$\Phi(\Omega, \Omega') = \delta(\Omega' - \Omega)$$
$$\rightarrow \quad \mathsf{S}_{n-1}(\tau) = a I_{n-1}(\tau)$$

- The iterative solutions are:



$$I_0(\tau) = e^{-\tau} I_0(0)$$
$$\rightarrow \quad \mathsf{S}_0(\tau) = a I_0(\tau) = a e^{-\tau} I_0(0)$$
$$I_1(\tau) = e^{-\tau} \int_0^\tau e^{\tau'} \mathsf{S}_0(\tau') d\tau' = (a\tau) e^{-\tau} I_0(0)$$
$$\rightarrow \quad \mathsf{S}_1(\tau) = a I_1(\tau) = (a^2 \tau) e^{-\tau} I_0(0)$$
$$I_2(\tau) = e^{-\tau} \int_0^\tau e^{\tau'} \mathsf{S}_1(\tau') d\tau' = \frac{(a\tau)^2}{2} I_0(0)$$
$$\rightarrow \quad \mathsf{S}_2(\tau) = a I_2(\tau) = (a^3 \tau^2) e^{-\tau} I_0(0)$$
$$I_3(\tau) = e^{-\tau} \int_0^\tau e^{\tau'} \mathsf{S}_2(\tau') d\tau' = \frac{(a\tau)^3}{3 \times 2} I_0(0)$$
$$\vdots$$
$$\rightarrow \quad \mathsf{S}_{n-1}(\tau) = a I_{n-1}(\tau) = (a^n \tau^{n-1}) e^{-\tau} I_0(0)$$
$$I_n(\tau) = e^{-\tau} \int_0^\tau e^{\tau'} \mathsf{S}_{n-1}(\tau') d\tau' = \frac{(a\tau)^n}{n!} I_0(0)$$

The final solutions are:

$$I^{\rm direc}(\tau) = e^{-\tau} I(0)$$
$$\begin{aligned} I^{\rm scatt}(\tau) &= \sum_{n=1}^{\infty} I_n(\tau) = \sum_{n=1}^{\infty} \frac{(a\tau)^n}{n!} e^{-\tau} I(0) \\ &= (e^{a\tau} - 1) e^{-\tau} I(0) \\ &\approx a\tau e^{-\tau} I(0) \quad \text{if} \quad a\tau \ll 1 \\ &\approx a\tau I(0) \qquad \text{if} \quad \tau \ll 1 \end{aligned}$$

$$\begin{aligned} I^{\rm tot}(\tau) &= I^{\rm direc}(\tau) + I^{\rm scatt}(\tau) \\ &= e^{-(1-a)\tau} I(0) \\ &= e^{-\tau_{\rm abs}} I(0) \end{aligned}$$